
The Geometry of Empowerment

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Abstract

Empowerment captures the capacity for an agent to actively control its environment. While conceptually appealing as an information-theoretic quantity, the connection between empowerment and MDP centrality has remained an open question. In this work, we link empowerment maximization and skill-learning methods to provide new geometries for interpreting and analyzing empowerment. Our analyses answer longstanding open questions on the connections between empowerment and centrality. Our analysis also reveals distinctions between information and reward geometries, highlighting important theoretical implications to build scalable empowerment-maximization methods. The code for the tabular experiments accompanying our theoretical results is at <https://github.com/cj7280/skill-simplex>.

1 Introduction

Two trained empowerment-maximizing policies characterize central states in an MDP. Consider an agent (robot) on a hill that is easier to descend than to ascend. (a) The potential policy ascends the empowerment landscape, using $\mathcal{E}_{\text{pot}}(s)$ as an intrinsic reward. (b) The effective policy π_{eff} maximizes $I^\pi(Z; S_+ | S_0)$, associating each skill z with a downhill route.

Figure 1: **Estimating empowerment requires two distinct policies: ...**

Two trained empowerment-maximizing policies characterize central states in an MDP. Consider an agent (robot) on a hill that is easier to descend than to ascend. (a) The potential policy ascends the empowerment landscape, using $\mathcal{E}_{\text{pot}}(s)$ as an intrinsic reward. (b) The effective policy π_{eff} maximizes $I^\pi(Z; S_+ | S_0)$, associating each skill z with a downhill route.

An intelligent agent should be in control of its future. For example, a well-trained hiker at a mountain can (1) choose and traverse down one of many trails from a mountain peak (§. 1(b)) or (2) navigate to and stay at the peak (§. 1(a)). While both useful, these behaviors are distinct: the first behavior characterizes “actively exerting control” while the latter characterizes “being in control.”

Cleanly parameterizing, optimizing, and separating these behaviors requires a tractable mathematical formulation of controllability. **Empowerment** is an information-theoretic object that quantifies controllability in terms of bits. As originally defined in [1], **potential empowerment** measures the maximal bits an

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agent’s actions can inform its future state from some current state. In other words, the potential empowerment measures how much control an agent *could* command from a given state. An alternative measure of empowerment, termed here as **effective empowerment**, is a mutual information that quantifies how readily an agent’s policy *actively* informs its future state [2]. While sometimes conflated, these two objects are different. In the example of the mountain in §. 1, a potential empowerment policy ascends the mountain to a state from which the agent has options. Meanwhile, the effective empowerment policies actively exercise options and descend the mountain in different directions.

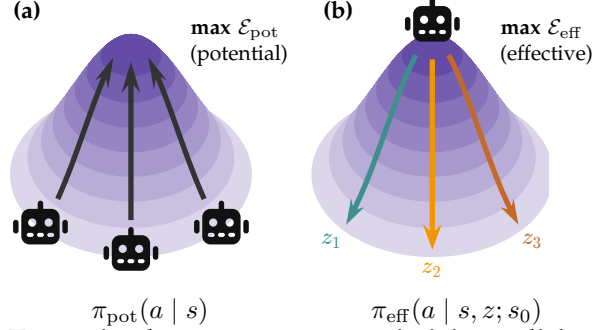
Empowered States are Central States. Prior works conjecture that empowerment is a generic way for agents to do goal-oriented tasks. Salge et al. [3, §4.3.3] call this the AI Empowerment Hypothesis:

Empowerment provides a task-independent motivation that generate AI behaviour which is beneficial for a range of goal-oriented behaviour.

However, empowerment does not have immediate obvious connections to downstream task performance. Indeed, while early works in empowerment suggest a link between empowerment and preferred or central states [3–5], to the best of our knowledge, formally characterizing notions of MDP centrality and adaptation in relation to empowerment has remained a longstanding open problem. In this work, we formally show high potential empowerment states lie at centers of reachable regions in the state occupancy simplex (Theorem 4.1). The empowerment objective and long horizon generalizations also admit exact hitting time interpretation (Lemma 4.2) and capture centrality in a temporal distance geometry [6]. In addition to providing intuition, these notions of centrality directly lower bound an agent’s ability to adapt to downstream tasks in tabular settings (Theorem F.1), providing a partial answer to Salge’s longstanding hypothesis. Thus, potentially empowered agents can reach central states from which they can adapt, where effective empowerment quantifies the adaptation.

Information Geometries and Reward Geometries can Diverge in Continuous Settings. The above results theoretically answer questions in empowerment literature: (1) empowered states are central states and (2) in discrete settings, empowered states are states from which agents can adapt to downstream tasks. Attempting to extend the adaptation result to continuous settings reveals that information and reward geometries can diverge: an arbitrarily empowered state can provide arbitrarily small adaptation in a continuous MDP with continuous rewards (Proposition G.2). The same misalignment appears in discrete settings once the reward prior is anisotropic (Proposition H.1). Thus, empowered agents are *not generally* adaptive agents in the continuous setting. Scalable formulations of empowerment maximization must address this discrepancy between information and reward adaptation geometries.

Taken together, our results give an information-geometric centrality theorem for the empowerment objective and a complementary temporal interpretation over skill rollouts. These theoretical interpretations provide an answer to longstanding questions about the relationship between empowerment, centrality, and adaptation to downstream reward functions: in tabular settings, empowered agents navigate to central states from which they can adapt. In continuous settings and under anisotropic rewards, this intuition no longer holds. Thus, our theoretical results show that scalably unifying empowerment and adaptation requires alternative formulations that align information and reward geometries.



Two trained empowerment-maximizing policies characterize central states in an MDP. Consider an agent (robot) on a hill that is easier to descend than to ascend. (a) The potential policy ascends the empowerment landscape, using $\mathcal{E}_{\text{pot}}(s)$ as an intrinsic reward. (b) The effective policy π_{eff} maximizes $I^\pi(Z; S_+ | S_0)$, associating each skill z with a downhill route.

2 Related Work

As originally introduced, the empowerment objective captures the maximum possible information that can be communicated from a sequence of open-loop actions to future states, provided the side information (“context”) of the current state [7]. This empowerment is a channel capacity defined at a particular context and state s . Thus, an empowerment maximizing agent should maximize the empowerment of its occupied states. Note that the optimization variable here is an open-loop action sequence, though closed-loop variants have been discussed [3, 8].

Later works term this channel capacity form of empowerment as *potential empowerment* [2]: an agent can become more empowered by reaching high potential empowerment *states*. Correspondingly, Capdepuy [2] defines a notion of *actualized empowerment* that is a property of a single closed-loop policy, measuring the mutual information between the actions chosen by the policy and a future state conditioned on the current state. In this work, we refer to this form of empowerment as *effective empowerment*. A policy has high effective empowerment if its actions readily communicate many bits to future observed states.

Where should an agent go if told that it will be asked to solve some soon-to-be-specified task? One approach is to consider the geometry of the state space and navigate to **central states** or **bottleneck states** within the MDP: states from which the agent can easily visit many other states [9]. Such notions of MDP centrality have been studied in both reinforcement learning [4, 5, 10] and neuroscience [11] community, often in the context of exploration. While these notions and definitions of centrality and bottlenecks are appealing, these objects are only well-defined in settings with discrete states and actions.

Prior empowerment literature has conjectured that potential empowerment measures the centrality of a state in an MDP [4, 5]. For example, Salge et al. [3] note that potential empowerment exactly corresponds to the log of the total reachable states in discrete, deterministic settings. In stochastic settings, prior works discuss the relationship between empowerment and closeness centrality and show correlations between these quantities in simple didactic experiments; however, no formal link between centrality and empowerment is provided. Jung et al. [5] mention the possibility that empowerment might identify “gateway” states, but do not explicitly test nor quantify this observation in experiments. Our work provides a rigorous grounding to these observations, formally connecting potential and effective empowerment to two different notions of centrality: centrality in temporal distance geometry and centrality in the information geometry.

One important consequence of our theoretical results is a notion of centrality that remains well defined in continuous settings. A natural generalization of discrete measures of reachability and closeness is the temporal distance [6]. In discrete settings, the temporal distance corresponds to a simple function of the hitting time from state-to-state. Previous work has shown temporal distances can effectively guide exploration [12] and goal-reaching [13, 14] in continuous, high-dimensional tasks. To our knowledge, there has not been prior work utilizing temporal distances to identify and characterize centralized or bottleneck states. Here, we show that empowerment corresponds to the reduction in time/temporal distance to future states.

Estimating and Maximizing Empowerment in Practice. There are two different forms of empowerment: potential empowerment and effective empowerment [2]. While prior works discuss maximizing both quantities [15, 16], to the best of our knowledge, there have been no prior works that scalably maximize potential empowerment in high-dimensional, continuous settings. Meanwhile, there has been substantially more prior work in RL estimating and maximizing *effective empowerment* (or similar) quantities in high-dimensional, continuous settings, in a line of work termed Mutual Information Skill Learning (MISL). Prior works use similar mutual information objectives as Mohamed and Jimenez Rezende [8] to learn a latent z -conditioned policy $\pi(a | s, z)$ that transmits bits about an input vector Z to future states S_+ via its actions (i.e., closed-loop control [16–19]):

$$\mathcal{E}_{\text{eff}}(\pi) = I^\pi(Z; S_+ | S),$$

which replaces the action A with a latent Z . These methods learn *skills* $\pi(a \mid s, z)$ that actively transmit information to the future, which are high effective empowerment policies.

By making use of contrastive and variational estimators of mutual information, prior works scale these methods to learn reactive policies in high-dimensional (e.g., with images) continuous-control tasks [18]. While some of these skill-learning methods mention empowerment [16], they do not directly optimize potential empowerment as defined in [1] and instead only optimize effective empowerment. Recently, empowerment and its variants have also served as objectives for assistance [20–22]. While we do not present experiments in high-dimensional or continuous settings, we highlight these lines of work to emphasize that empowerment and related objectives underpin state-of-the-art intrinsic rewards for unsupervised reinforcement learning (RL) and new assistance methods. Understanding the theoretical reasons to maximize or modify empowerment therefore directly informs the design of these important methods.

3 Preliminaries

3.1 Markov Decision Process (MDP) and Discounted State Occupancy Preliminaries

MDP problem setting. Our analysis focuses on applying information theory to a fully-observed MDP with states $s \in \mathcal{S}$, actions $a \in \mathcal{A}$, and environment dynamics $p(s' \mid s, a)$. Let the skill and starting state-conditioned policy be the function $\pi(a \mid s, z; s_0)$ with skill $z \in \mathcal{Z}$.

The latent skill z lives in a continuous skill space $\mathcal{Z} \subseteq \mathbb{R}^d$ (i.e. the surface of a hypersphere), including in tabular MDPs. The policy $\pi(a \mid s, z; s_0)$ is a distribution over actions given the current state s , the latent z , and the start state s_0 . We write $\Pi_{\mathcal{Z}}$ for this class of skill- and starting-state conditioned policies, and Π for the class of ordinary policies $\pi(a \mid s)$ that carry no skill. Fixing a skill in a policy of $\Pi_{\mathcal{Z}}$ leaves an ordinary policy in Π . The start-state s_0 is the context of a skill [7].

Reachable state-occupancy polytope. For some initial state s_0 , skill latent $z \in \mathcal{Z}$, and skill and start-state-conditioned policy $\pi(a \mid s, z; s_0)$, the discounted state occupancy measure (DSOM) takes the form

$$p^\pi(s_+ = s \mid s_0, z) \triangleq (1 - \gamma) \sum_{t=0}^{\infty} \gamma^t p^\pi(s_t = s \mid s_0, z), \quad (1)$$

where $p^\pi(s_t = s \mid s_0, z)$ is the probability that policy π visits state s after t time steps when initialized at state s_0 under skill z .

Averaging over the skill distribution $p(z \mid s_0)$ gives the skill-marginal DSOM,

$$p^\pi(s_+ \mid s_0) = \mathbb{E}_{p(z \mid s_0)} [p^\pi(s_+ \mid s_0, z)], \quad (2)$$

the future-state distribution induced by the policy when the skill is unknown.

Every DSOM is a distribution over states within the simplex $\Delta(\mathcal{S})$. For a fixed MDP, an agent can only realize a strict subset of DSOM in the simplex. For a fixed start state s_0 , the realizable DSOMs form the *reachable state-occupancy polytope*

$$\mathcal{P}(s_0) \triangleq \{ p^\pi(s_+ \mid s_0) : \pi \in \Pi \} \subseteq \Delta(\mathcal{S}), \quad (3)$$

a convex subset of the simplex whose linear constraints are the Bellman flow equations (see the polytope inside the simplex in Fig. 3a) [23]. We write $\mathcal{P}(s)$ for the polytope reachable from a generic start state s . Our analysis in §. 4 studies the geometry of exactly this polytope.

3.2 Empowerment Preliminaries

Potential and Effective Empowerment. Our work highlights the difference between empowerment, as originally defined in Klyubin et al. [24] and the mutual information (MI) objects directly maximized in

MISL methods [8, 17–19]. We term these forms of empowerment as potential and effective empowerment, respectively. The distinction between these forms of empowerment is not new and has been pointed out in prior work [2], though (to our knowledge) modern works do not always differentiate between these objects [8].

Definition 3.1 (Potential empowerment). Let policies $\pi \in \Pi_{\mathcal{Z}}$ condition on skill latents $z \in \mathcal{Z}$ and let S_+ denote the random state $s_+ \sim p^\pi(s_+ | s_0, z)$ sampled from the DSOM. Potential empowerment is the channel capacity over a *family of policies* dependent on an initial state [1, 2]

$$\mathcal{E}_{\text{pot}}(s_0) = \max_{\pi \in \Pi_{\mathcal{Z}}} I^\pi(Z; S_+ | S_0 = s_0).$$

Definition 3.2 (Potential empowerment policy). A potential empowerment policy $\pi_{\text{pot}} \in \Pi$ that takes the form $\pi_{\text{pot}}(a | s)$ maximizes the potential empowerment as a reward:

$$\pi_{\text{pot}}^* = \arg \max_{\pi \in \Pi} \mathbb{E}_\pi \left[\sum_t \gamma^t \mathcal{E}_{\text{pot}}(s_t) \right].$$

While conceptually similar, mutual information skill learning methods do not maximize potential empowerment, and maximize effective empowerment. *Effective empowerment* captures the MI itself:

Definition 3.3 (Effective empowerment). Consider skill and start state-conditioned policies $\pi \in \Pi_{\mathcal{Z}}$ that take the form $\pi(a | s, z; s_0)$. *Effective empowerment* is the mutual information realized by a particular skill and start state-conditioned policy at a particular starting state s_0 under the uniform skill source $p(z | s_0) = \text{Unif}(\mathcal{Z})$:

$$\mathcal{E}_{\text{eff}}(\pi, s_0) = I^\pi(Z; S_+ | S_0 = s_0) = \mathbb{E}_{p(z|s_0)} D_{\text{KL}}[p^\pi(s_+ | s_0, z) \| p^\pi(s_+ | s_0)]$$

Correspondingly, we refer to the skill and start state-conditioned policy $\pi(a | s, z; s_0)$ as the *effective empowerment policy*.

This objective is similar to the objectives directly maximized in modern skill-learning algorithms [16–18]; note that MISL methods generally assume a fixed starting state s_0 , so the learned skills are specific to s_0 . Thus, we also refer to the effective empowerment policies as skills.

Importantly, effective empowerment involves no maximization and measures the MI *achieved* by a particular skill-conditioned policy $\pi \in \Pi_{\mathcal{Z}}$ that begins from a particular starting state s_0 . This MI depends on both the *channel* $z \mapsto p^\pi(s_+ | s_0, z)$, determined by the policy, and the *source* $p(z | s_0)$ set to uniform. Potential empowerment is the maximum-achievable effective empowerment for a given starting state s_0 , so that

$$\mathcal{E}_{\text{pot}}(s_0) = \max_{\pi \in \Pi_{\mathcal{Z}}} \mathcal{E}_{\text{eff}}(\pi, s_0) = \max_{\pi \in \Pi_{\mathcal{Z}}} \max_{p(z|s_0)} I^\pi(Z; S_+ | S_0 = s_0). \quad (4)$$

We can write the second equality because the source is overparameterized. Under a uniform skill prior, the policy can represent any skill source by simply relabeling skills (Lemma D.1). Fixing the uniform source in Definition 3.1 therefore loses no generality, with the full statement in Appendix D.

Thus, a learned skill set (effective empowerment policies) acts as a probe of the potential empowerment at a starting state. If skills initialized at s_0 induce easily distinguishable future occupancies $p^\pi(s_+ | s_0, z)$, then s_0 affords many controllable futures and has high potential empowerment $\mathcal{E}_{\text{pot}}(s_0)$.

Notation convention. Throughout the paper, we write π for the effective policy when context is clear and π_{eff} only when both policies appear in the same expression. The potential policy is always written as π_{pot} . The object $\mathcal{E}(s)$ always refers to potential empowerment measured at starting-state s .

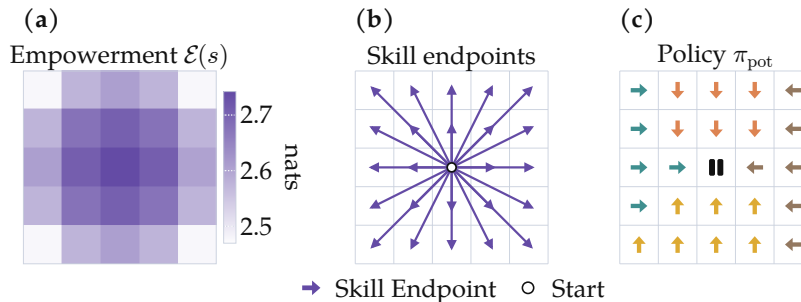


Figure 2: **Skills measure the centrality of a state.** (a) We visualize the potential empowerment field $\mathcal{E}(s)$ for each state in a tabular GridWorld. The central state has high potential empowerment, because (b) the central state gives an agent access to a maximal set of distinguishable skills. (c) A potential empowerment policy ascends the potential empowerment landscape to the center state, from which it can execute different skills. See Theorem 4.1 for the formal statement, see Appendix K for tabular experiment details.

4 The Geometry of Empowerment

Prior work has empirically observed a close connection between geometry and empowerment [4, 5], wherein states that are more centralized in an MDP have higher empowerment. However, to the best of our knowledge, such connections have not been proven theoretically. In this section, we show that empowerment is closely related to a probabilistic notion of centrality in information geometry [25] and temporal distance geometry [6]:

- §. 4.1: Effective empowerment corresponds to the information-theoretic distance between states a *specific* skill-conditioned policy may reach in the future. Potential empowerment is the maximum over these distances from a starting state and thus measures the radius of a reachable space.
- §. 4.2: High empowerment states are central in temporal distance geometry. From a current state, the potential empowerment measures the average time saved to reach future states when committing to a skill.

We then connect empowerment to adaptation on downstream tasks (§. 4.3). We show a positive result, a lower bound on adaptation, but also prove important limitations that highlight the fundamental difference between information and reward geometries:

- §. 4.3, Theorem F.1: The potential empowerment of a state lower bounds adaptation to unknown reward functions drawn from an isotropic distribution.
- §. 4.3, Proposition G.2: In continuous settings, a state can have unbounded potential empowerment while providing arbitrarily small adaptation to a continuous reward.
- §. 4.3, Proposition H.2: For anisotropic reward distributions, an empowered state can provide no adaptation.

4.1 Information Geometry of Potential and Effective Empowerment Policies.

Geometric intuition. Prior work has found empirically that high empowerment states are central states in an MDP, relating empowerment to the volume of reachable states in discrete, deterministic settings [3]. Here, we extend definitions of centrality to the full MDP setting with stochasticity and continuous states.

Effective empowerment is a mutual information. In the dual form, effective empowerment maximization finds state occupancy distributions that lie at the center of extremized state distributions. Then, potential

empowerment-seeking policies search for the states that lie at the centers of the largest radii reachable polytopes.

Before presenting formal results, we visualize an example of the information geometry of empowerment in Fig. 3. Let the triangle visualize the state occupancy simplex reachable from different states s_1 and s_2 . Maximizing the mutual information (effective empowerment) for a given starting state corresponds to learning skills that reach vertices of the set of reachable state occupancies. The agent maximizes effective empowerment separately at every possible starting state (yellow for s_1 , blue for s_2) and learns skills (yellow and blue arrows) that span reachable regions. Then, to maximize potential empowerment, an agent searches among these starting states for the largest reachable region. In this case, the blue state s_2 corresponds to the state with the highest potential empowerment and occupies a more centralized point within the information geometry.

High Empowerment States are Central in Information Geometry. Formally, consider the state probability simplex $\Delta(\mathcal{S})$, where DSOM are elements of the simplex $p^\pi(s_+ | s_0, z) \in \Delta(\mathcal{S})$. A straightforward extension of prior work [23] shows that effective empowerment maximization is a center localization problem in the initial state s_0 -conditioned state probability simplex, where distance is given by the Kullback–Leibler divergence [26]. Then, potential empowerment measures the min-max distance to the reachable state-occupancy polytope:

Theorem 4.1 (Potential empowerment is the Kullback-Leibler (KL) radius of the reachable polytope, extending Eysenbach et al. [23, Lem. 6.1–6.2]). *Let $\mathcal{P}(s_0) \subset \Delta(\mathcal{S})$ be the reachable state-occupancy polytope of Eq. (3). Jointly maximizing the mutual information over the skill source and the skill-conditioned policy solves the KL 1-center localization problem for $\mathcal{P}(s_0)$:*

$$\mathcal{E}_{\text{pot}}(s_0) = \max_{\pi \in \Pi_{\mathcal{Z}}, p(z|s_0)} I^\pi(\mathcal{Z}; S_+ | S_0 = s_0) = \min_{q(s_+) \in \Delta(\mathcal{S})} \max_{p \in \mathcal{P}(s_0)} D_{\text{KL}}(p(s_+) \| q(s_+)). \quad (5)$$

Moreover, every joint optimum $(\pi, p(z | s_0))$ places every skill on the boundary of the optimal KL ball measured from the optimal central occupancy q^ :*

$$D_{\text{KL}}(p^\pi(s_+ | s_0, z) \| q^*(s_+)) = \max_{p \in \mathcal{P}(s_0)} D_{\text{KL}}(p(s_+) \| q^*(s_+)) \quad \text{for every skill } z. \quad (6)$$

In other words, the optimized skills span the maximal information radius of the reachable polytope given starting state s_0 .

We prove a general version of Theorem 4.1 in Appendix B. This result shows that an exploration agent maximizing potential empowerment will seek central states in the information geometry. Prior work on empowerment was correct in conjecturing that empowerment maximization has a formal connection to centrality. However, the right geometry for studying this centrality is not (say) the Euclidean metric on states, but rather the KL divergence between state distributions induced by different skills.

A simple tabular experiment demonstrates this notion of centrality (Fig. 3b). In this MDP, the agent can pick up and drop off a key at the the bottom of the hallway in addition to {Left, Right, Up, Down, Stay} actions. Picking up this key allows access to the blue-shaded regions of the MDP; without the key, the agent cannot enter these spaces in the grid. The potential empowerment policy π_{pot} that ascends the potential empowerment landscape correctly reaches the true central state: the center of the MDP with the key. This experiment shows that states with maximal potential empowerment capture an information-geometric notion of centrality, rather than navigating towards the *physically* central state.

4.2 Empowerment, Bottlenecks, and Temporal Distance

Beyond information geometry, the empowerment objective has a direct hitting time interpretation in discrete state spaces. Suppose that the agent has multiple trial trajectories to reach a future state, where each trajectory rolls out a skill from the same starting state until a geometric termination time. The empowerment measures the reduction in hitting time when the agent commits to a skill:

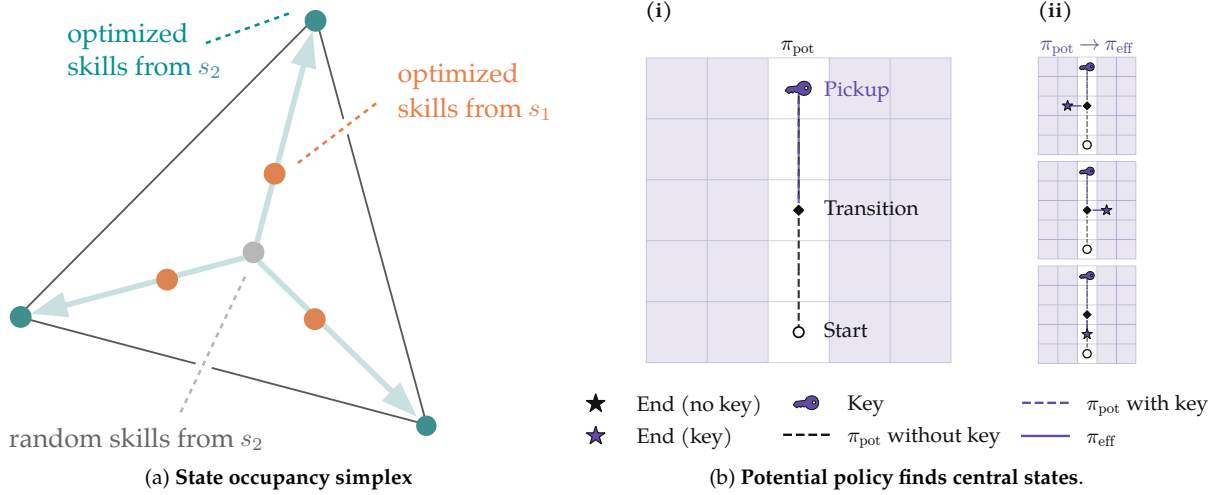


Figure 3: **Potentially empowered states are central states in the polytope.** Tabular intuition for Theorem 4.1. (a) We can use a probability simplex $\Delta(S)$ to visualize the discounted state occupancy measures of each skill (circles). Orange circles indicate the best set of skills starting from state s_1 , and green circles indicate the best skills starting from state s_2 . From starting states s_1 and s_2 in arbitrary MDP. The skills starting from s_1 define a smaller polytope (yellow points) in the simplex than s_2 (blue points) as a result of their transition dynamics, so the potential empowerment is higher at s_2 than s_1 . (b) We use a GridWorld to illustrate the connection between centrality and tool use. An agent in the GridWorld begins at the bottom of the hallway. The agent can set down and pick up a key at the top of the hallway. The key provides the agent access to shaded regions of the MDP. (b, i) A potentially empowered agent picks up the key then navigates to the center of the room, correctly identifying the middle state *with the key* as the most central state. (b, ii) From the center, the agent can execute the maximal number of distinguishable skills. See Appendix K.3 for experimental details.

Lemma 4.2 (Bottleneck states are high potential empowerment states). *In a discrete MDP, let the hitting time random variable $H_{\text{rst}}^{\pi_{\text{eff}}}(s_0 \rightarrow s_+)$ be the number of rollouts from s_0 until a rollout hits s_+ , for skills $Z \sim p(z | s_0)$. Let $H_{\text{rst}}^{\pi_{\text{eff}}}(s_0 \rightarrow s_+ | z)$ be the corresponding hitting time when rolling out z every time. Then,*

$$\mathcal{E}_{\text{pot}}(s_0) = \max_{\pi_{\text{eff}} \in \Pi_Z} \mathbb{E}_{p^{\pi_{\text{eff}}}(z, s_+ | s_0)} [\log \mathbb{E}[H_{\text{rst}}^{\pi_{\text{eff}}}(s_0 \rightarrow s_+)] - \log \mathbb{E}[H_{\text{rst}}^{\pi_{\text{eff}}}(s_0 \rightarrow s_+ | z)]] .$$

We prove Lemma 4.2 in Appendix C. In tabular settings, potential empowerment measures the reduction in time to reach future states *over independent rollouts* when committing to an option/skill.

However, hitting times are not well-defined in continuous geometries, where hitting a particular state is a measure-zero event. The temporal distance is an object that remains well-defined in continuous settings and reduces to hitting times in the discrete limit [6]. In the appendix, we show that a closely related empowerment objective over *contiguous* (versus repeated) skill rollouts, where the agent re-samples a skill at each geometric termination in a single trajectory, admits an exact temporal distance formulation [6]. In that construction, the empowerment object measures the reduction in temporal distance to future states after committing to a skill (Proposition C.1), recovering a hitting-time interpretation in the discrete, deterministic limit.

Conceptually, these temporal formulations capture the bottleneck effect. From a bottleneck, an agent must commit to different skills in order to quickly reach different downstream regions which is conceptually the reduction in temporal distance and hitting times considered in our results. Tabular experiments in Fig. 4 illustrate this effect [11, 27]. Adding slim openings (“bottlenecks”) to the grid shifts the high empowerment states from the original central state to the bottleneck openings, prioritizing states where committing to a skill (e.g. move left or move right) quickly reaches different regions. This interpretation also connects empowerment to closeness centrality [4].

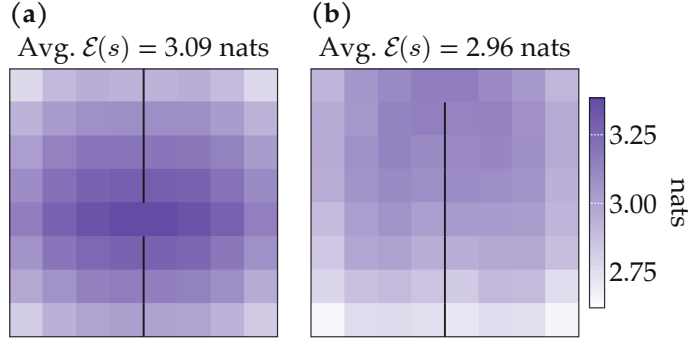


Figure 4: **Bottleneck states are high potential empowerment states.** Using a gridworld with a door in the middle, we visualize how shifting the door position shifts high-empowerment states from the original central state to the bottleneck openings. In §. 4.2 we relate empowerment to hitting times and temporal distance. See Appendix K.2 for experimental details.

4.3 Is Empowerment Optimal for Downstream Adaptation?

Potential empowerment lower bounds adaptation in stochastic discrete MDPs. While empowerment is often introduced as an appealing “universal” [24] objective, is there a precise sense in which empowerment measures the ability to adapt? In Appendix F, we formally prove that empowerment lower bounds skill adaptation to arbitrary rewards.

Because there are no target rewards during training, we must define an adaptation objective with respect to a reward prior distribution $\mathcal{R}$. A simple reward prior is a Gaussian distribution centered at a fixed baseline distribution, set to the uniform occupancy over the state space. Prior works use similar isotropic reward distributions when proving general downstream adaptation results [28].

Here, we define adaptation as a *skill adaptation* to a sampled reward. At test time, an agent must select a skill $z \in \mathcal{Z}$ to adapt to a sampled reward function $r \sim \mathcal{R}$. For example, in the case of the 5×5 grid, an agent should select the skill “go left” if the reward provided at test time is on the left.

In the best case, the agent selects the skill z with the highest expected return; note that the agent can do no better within the space of learned skills. This type of policy selection for adaptation/transfer has precedent in prior successor feature literature [29]. For the following result, we consider linear adaptation objective J for the skill set learned by an empowerment-maximizing agent, which is exactly the best skill’s mean return averaged over the reward prior. This adaptation objective is notably different from prior results connecting MI and adaptation in Eysenbach et al. [23], which considers regularized adaptation directly in the state occupancy. Here, the adaptation is the selection of a *pre-trained* skill behavior, which does not assume the ability to directly adapt state occupancy distributions:

Theorem 4.3 (Informal statement that tabular empowerment lower bounds reward adaptation). *Fix a start state s_0 and sample a reward $r \sim \mathcal{R}$ from the uninformed Gaussian prior. The agent adapts by selecting the skill with the highest mean return under r , and the adaptation $J(s_0)$ averages the best return over the reward prior. In whitened coordinates, write g for the reward and u_z for the offset of skill z ’s occupancy from the skill-marginal DSOM of Eq. (2), so the return gap of skill z over the marginal is $\langle g, u_z \rangle$. Then, for any skill space and skill prior,*

$$J(s_0) = \mathbb{E}_g \sup_{z \in \mathcal{Z}} \langle g, u_z \rangle \gtrsim I(Z; S_+ | S_0 = s_0).$$

We give the formal statement and proof in Theorem F.1 (Appendix F), where $J(s_0)$ turns out to be the Gaussian width of the skill offsets $\{u_z\}_{z \in \mathcal{Z}}$ (Eq. (12)). Intuitively, for the simple 5×5 grid (Fig. 2), the central maximal potential empowerment state is *exactly* the state from which an agent can easily adapt to the left/right/upper/lower rewarding states in the grid. Thus, maximizing potential empowerment provides a new form of provable adaptation where agents navigate to central states *from which* they can

later adapt, which existing skill-learning or effective empowerment methods do not capture on their own [16].

Two didactic tabular experiments demonstrate the behavior of the bound across MDP parameters. The first set of MDPs varies the degree of commitment within an MDP, which measures how much probability mass an action places on a target state above uniform. Commitment measures the controllability of the MDP. As controllability increases, the bound tightens (Fig. 10). The second set of MDPs varies the number of possible distinct skills/actions while holding commitment fixed. As the number of possible skills grows, the bound loosens (Fig. 11). The bound is therefore most informative in MDPs where an agent exerts strong control over comparatively few distinguishable regions. Full experimental details are in Appendix F.

The First Limitation of Empowered Adaptation: Gaussian widths versus KL divergence. While the above result demonstrates that potential empowerment lower bounds adaptation in discrete MDPs, there remain open questions: (1) is the bound tight? and, correspondingly, (2) does the bound extend to continuous geometries? The answer to both of these questions is generally no: without additional assumptions, the geometry of *reward adaptation* and that of *information geometry* are different.

First, we must define the geometry of reward adaptation. *Gaussian width* measures the spread of the skills’ future occupancies and governs the adaptation geometry [25]. Because the cumulative reward is *linear* in the occupancy, the width is large when different skills reach futures with clearly different values (Fig. 5, (b,e)). This quantity differs from MI, where distances are given by the Kullback–Leibler divergence.

Our first observation is that the adaptation result is a bound, not a ranking result. Consider the simple example of two starting states in a discrete MDP (Fig. 5, (a) and (d)). The first starting state in (a) can reach two distinct absorbing blocks. The second starting state (d), on the other hand, can reach three blocks.

Under an isotropic reward prior, an agent optimizing adaptation should choose the second starting state (Fig. 5, (e)) according to the Gaussian-width geometry that characterizes reward adaptation. Meanwhile, an agent optimizing the information radius will choose the first starting state (Fig. 5, (c)). This discrepancy is a direct consequence of the difference between information and reward adaptation geometry. Thus, while the lower bound holds, the bound does not guarantee a ranking result with provable counterexamples.

In the continuous limit, the difference between empowerment and adaptation geometry becomes more salient:

Theorem 4.4 (Informal statement that large MI does not necessarily imply adaptation to Lipschitz-continuous rewards in continuous MDPs). *For every skill count K and ϵ , there exists a continuous MDP admitting K distinct skills that induce DSOMs within an ϵ -ball in arbitrary continuous space.*

Taking $K \rightarrow \infty$, the MDP admits a set of skills that induce unbounded empowerment while adaptation to a Lipschitz-continuous reward can be arbitrarily small.

We give the formal statements and proofs in Propositions G.1 and G.2. Our result shows that skills can have unbounded distinguishability but fail to adapt to Lipschitz-continuous rewards, inducing identical returns across the skill set Fig. 6 illustrates how information geometries can lead to collapsed returns in a continuous 2D state space, where the potential empowerment grows from left to right while returns homogenize. Thus, larger potential empowerment does not imply better adaptation to smooth rewards in continuous settings. We fully discuss this collapse in Appendix G.

The Second Limitation of Empowered Adaptation: Controllable Degrees of Freedom versus Rewarding Degrees of Freedom. The second limitation of empowered adaptation is the misalignment between controllable degrees of freedom and preferred degrees of freedom. While the adaptation result of Theorem F.1 requires an isotropic reward prior, naive empowerment maximization cannot generically provide adaptation guarantees for anisotropic reward priors. Without external information,

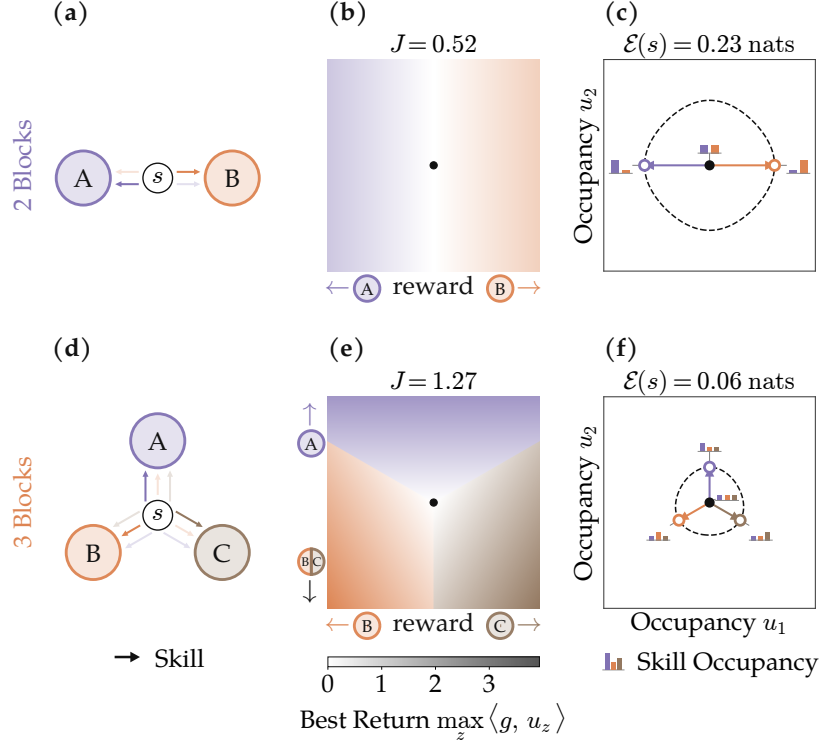


Figure 5: **Discrepancy between information geometry and reward adaptation geometry in discrete settings.** While we prove that potential empowerment lower bounds adaptation in tabular settings, a starting state with higher empowerment does not guarantee better adaptation (i.e. the bound is valid but is not a ranking result). Consider the starting state in (a) that admits two distinct skills reaching two absorbing blocks, while the state in (d) admits three overlapping skills that target three absorbing blocks. (b, e) visualizes the return gap as a function of the whitened reward $g = (g_1, g_2)$. In (e), larger g_1 favors state C over B , while larger g_2 favors A over states B/C . The saturation of the field at g denotes the adapted skill's return gap at that reward. Color indicates the adapted skill $\arg \max_z \langle g, u_z \rangle$. (c, f) show the information geometry of the skill occupancies. The dashed line indicates the level set of the maximal MI. Even though empowerment is higher for the 2 block state versus the 3 block state, the 2 block state has worse adapted returns. See Appendix K.4 for details.

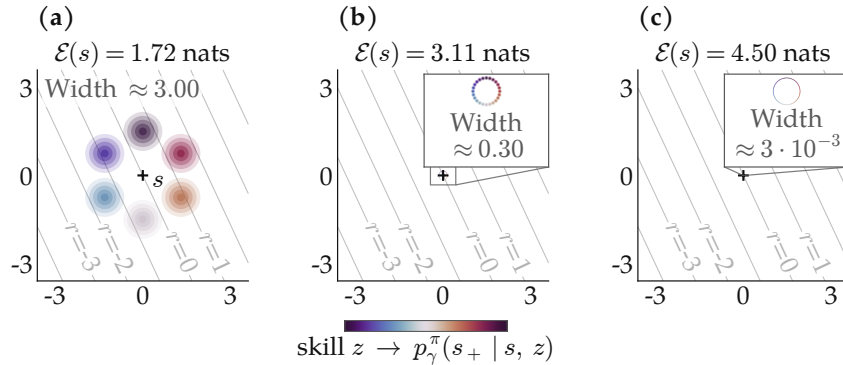


Figure 6: **The information ball can collapse in the continuous limit** (Proposition G.2). Consider an arbitrary Lipschitz-continuous reward function (level curves in gray) and some MDP that admits Gaussian skill state occupancy measures (colored distributions). While the potential empowerment of the starting state $s \in \mathcal{S}$ increases from (a) $\rightarrow$ (c), the skills have increasingly identical returns and, thus, provide progressively smaller adaptation benefit. Therefore, large channel capacities do not imply adaptation for continuous MDPs and reward functions. See Proposition G.2 for the theoretical result.

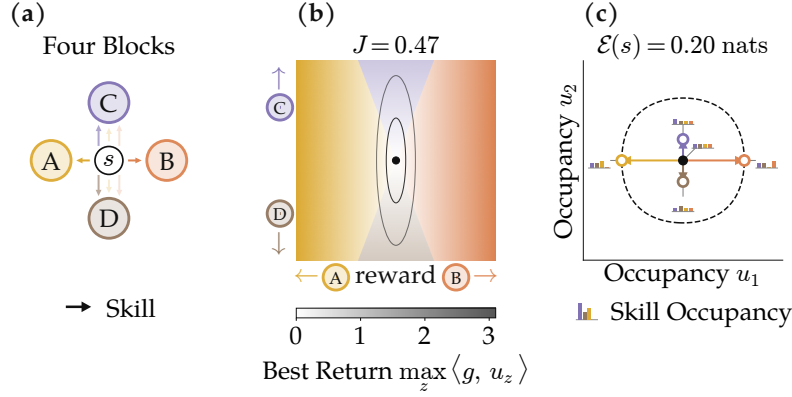


Figure 7: **Empowering skills do not guarantee adaptation under an anisotropic reward** (Proposition H.1). (a) Let A/B/C/D be absorbing blocks of states. In this four-block MDP, the left/right skills commit strongly to their blocks A/B, while the up/down skills commit weakly to C/D; the arrow opacity denotes transition probability. (b) visualizes the adapted return gap as a function of the whitened reward $g = (g_1, g_2)$. Larger g_1 favors reaching B over A, while larger g_2 favors reaching C over D. The saturation of the field at g denotes the adapted skill’s return gap and color indicates the adapted skill. The solid ellipses are the 1σ and 2σ level sets of the anisotropic reward prior; the return depends mostly on how often an agent reaches C/D and is largely independent of how often an agent reaches A/B. However, (c) the skills reaching A/B are more distinguishable and have larger MI than the skills reaching C/D. Thus, due to the anisotropy of the reward prior, the skills that maximize the channel capacity do not provide optimal adaptation. See Proposition H.1 for the theoretical statement and Appendix K.4 for tabular experiment details.

empowerment can only break symmetry across states via controllability. Then, the reward prior is free to vary in directions in the state occupancy simplex that empowerment does not naturally select:

Theorem 4.5 (Informal statement that large MI need not imply adaptation to anisotropic reward function priors in discrete settings). *A state can have arbitrarily large MI while adaptation with respect to an anisotropic reward prior is arbitrarily close to zero.*

The formal statements and proofs are in Appendix H (Propositions H.1 and H.2). Fig. 7 visualizes an example where the maximizing MI selects for skills that do not control degrees of freedom relevant to the anisotropic return. Thus, the skill set is suboptimal for adaptation to the anisotropic reward prior.

5 Discussion, Limitations, & Future Work

In this work, we analyze the geometry of empowerment. Our theoretical results show that high potential empowerment states are central states in information geometry (Theorem 4.1). The empowerment objective and a multistep skill rollout variant also admit exact hitting time interpretations and capture centrality in a temporal distance geometry (Lemma 4.2 and Proposition C.1). Thus, policies that maximize potential empowerment naturally seek out central states within an MDP. Furthermore, effective empowerment policies, or skills, measure the degree of centrality (Fig. 2). Finally, in the tabular setting, this notion of centrality achieved by a set of skills directly lower bounds an agent’s ability to adapt to unknown rewards (sampled from an uninformed prior). Our results formally link empowerment and the ability to complete downstream tasks (Theorem F.1).

Limitations & Future Work. That being said, while the above results provide support for empowerment as a generic objective, information geometry and return geometries are fundamentally different: reward adaptation geometry is typically that of Gaussian widths, while information geometry is that of KL-divergences. This observation has a few downstream implications. First, a skill set that obtains a larger MI does not necessarily exhibit better adaptation; our adaptation result is a lower bound, not a ranking result (Fig. 5). Second, large empowerment does not imply adaptation in continuous MDPs with Lipschitz-continuous rewards: there exist MDPs where skill sets can obtain unbounded empowerment, while providing no adaptation benefit to a continuous reward function (Proposition G.2 and Fig. 6).

Third, empowerment *alone* does not provide theoretical guarantees on downstream adaptation to anisotropic reward priors (Proposition H.1) that prefer certain states over others.

Thus, while empowerment has attractive properties in addition to a straightforward information-theoretic interpretation, there exist theoretical gaps between the empowerment object and downstream task adaptation. Our work provides a partial answer to Salge’s hypothesis [3]: while empowerment, a task-agnostic quantity, does provably lower bound goal-oriented behavior in tabular settings, this hypothesis is not generally true for current formulations of empowerment maximization and return maximization. Possible future directions include developing scalable formulations of continuous empowerment maximization studied in prior works [5, 30, 31] and modifying recent scalable skill-learning algorithms for empowerment estimation [15, 18, 19].

Until we consider exploration and adaptation methods that appropriately unify these geometries, there will remain a disconnect between such intrinsic reward approaches and the broader RL problem setting. Indeed, our analysis can generically apply to any method that combines extrinsic and information-theoretic intrinsic rewards [32]. We believe that scalably addressing these limitations is key in order to make information-based intrinsic rewards, including empowerment, relevant to the broader RL community and is the natural next step for future work.

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References

- [1] A.S. Klyubin, D. Polani, and C.L. Nehaniv. Empowerment: A Universal Agent-Centric Measure of Control. *IEEE Congress on Evolutionary Computation*, volume 1, pp. 128–135 Vol.1, 2005.
- [2] Philippe Capdepu. Informational Principles of Perception-Action Loops and Collective Behaviours. 2011.
- [3] Christoph Salge, Cornelius Glackin, and Daniel Polani. Empowerment—an Introduction. *Guided Self-Organization: Inception*, pp. 67–114. Springer, 2014.
- [4] Tom Anthony, Daniel Polani, and Chrystopher L Nehaniv. On Preferred States of Agents: How Global Structure Is Reflected in Local Structure. 2008.
- [5] Tobias Jung, Daniel Polani, and Peter Stone. Empowerment for Continuous Agent—Environment Systems. *Adaptive Behavior*, 19(1):16–39, 2011.
- [6] Vivek Myers, Chongyi Zheng, Anca Dragan, Sergey Levine, and Benjamin Eysenbach. Learning Temporal Distances: Contrastive Successor Features Can Provide a Metric Structure for Decision-Making. *International Conference on Machine Learning*, 2024.
- [7] Alexander S Klyubin, Daniel Polani, and Chrystopher L Nehaniv. Keep Your Options Open: An Information-Based Driving Principle for Sensorimotor Systems. *PloS One*, 3(12):e4018, 2008.
- [8] Shakir Mohamed and Danilo Jimenez Rezende. Variational Information Maximisation for Intrinsically Motivated Reinforcement Learning. *Neural Information Processing Systems*, 28, 2015.
- [9] Gert Sabidussi. The Centrality Index of a Graph. *Psychometrika*, 31(4):581–603, 1966.
- [10] Özgür Şimşek and Andrew G Barto. Betweenness Centrality as a Basis for Forming Skills. 2007.
- [11] Kimberly L Stachenfeld, Matthew M Botvinick, and Samuel J Gershman. Design Principles of the Hippocampal Cognitive Map. *Neural Information Processing Systems*, 27, 2014.
- [12] Yuhua Jiang, Qihan Liu, Yiqin Yang, Xiaoteng Ma, Dianyu Zhong, Hao Hu, Jun Yang, Bin Liang, Bo Xu, Chongjie Zhang, and Qianchuan Zhao. Episodic Novelty Through Temporal Distance. arXiv:2501.15418, 2025.

- [13] Vivek Myers, Bill Chunyuan Zheng, Benjamin Eysenbach, and Sergey Levine. Offline Goal-Conditioned Reinforcement Learning With Quasimetric Representations. *arXiv:2509.20478*, 2025.
- [14] Seohong Park, Deepinder Mann, and Sergey Levine. Dual Goal Representations. *arXiv:2510.06714*, 2026.
- [15] Andrew Levy, Alessandro G Allievi, and George Konidaris. Representation Learning and Skill Discovery With Empowerment. *Reinforcement Learning Conference*, 2025.
- [16] Karol Gregor, Danilo Jimenez Rezende, and Daan Wierstra. Variational Intrinsic Control. *International Conference on Learning Representations*, 2016.
- [17] Benjamin Eysenbach, Abhishek Gupta, Julian Ibarz, and Sergey Levine. Diversity Is All You Need: Learning Skills Without a Reward Function. *International Conference on Learning Representations*, 2018.
- [18] Chongyi Zheng, Jens Tuyls, Joanne Peng, and Benjamin Eysenbach. Can a MISL Fly? Analysis and Ingredients for Mutual Information Skill Learning. *ICLR 2025*, 2025.
- [19] Seohong Park, Oleh Rybkin, and Sergey Levine. Metra: Scalable Unsupervised RL With Metric-Aware Abstraction. *International Conference on Learning Representations*, 2023.
- [20] Vivek Myers, Evan Ellis, Sergey Levine, Benjamin Eysenbach, and Anca Dragan. Learning to Assist Humans Without Inferring Rewards. *Neural Information Processing Systems*, 2024.
- [21] Evan Ellis, Vivek Myers, Jens Tuyls, Sergey Levine, Anca Dragan, and Benjamin Eysenbach. Training LLM Agents to Empower Humans. *International Conference on Machine Learning*, *arXiv:2510.13709*, 2026.
- [22] Claire Yang, Claire Jie Zhang, Maya Cakmak, and Max Kleiman-Weiner. When Assisting One Disempowers Another. *arXiv:2511.04177*, 2026.
- [23] Benjamin Eysenbach, Ruslan Salakhutdinov, and Sergey Levine. The Information Geometry of Unsupervised Reinforcement Learning. *International Conference on Learning Representations*, 2021.
- [24] Alexander S Klyubin, Daniel Polani, and Chrystopher L Nehaniv. All Else Being Equal Be Empowered. *European Conference on Artificial Life*, pp. 744–753, 2005.
- [25] Yury Polyanskiy and Yihong Wu. *Information Theory: From Coding to Learning*. Cambridge University Press, first edition, 2022.
- [26] Nimrod Megiddo. The Weighted Euclidean 1-Center Problem. *Mathematics of Operations Research*, 8(4):498–504, 1983.
- [27] Özgür Şimşek, Alicia P Wolfe, and Andrew G Barto. Identifying Useful Subgoals in Reinforcement Learning by Local Graph Partitioning. *International Conference on Machine Learning*, pp. 816–823, 2005.
- [28] Yann Ollivier. Which Features Are Best for Successor Features? *arXiv:2502.10790*, 2025.
- [29] André Barreto, Will Dabney, Rémi Munos, Jonathan J Hunt, Tom Schaul, Hado P van Hasselt, and David Silver. Successor Features for Transfer in Reinforcement Learning. *Neural Information Processing Systems*, 30, 2017.
- [30] Stas Tiomkin, Daniel Polani, and Naftali Tishby. Control Capacity of Partially Observable Dynamic Systems in Continuous Time. *arXiv:1701.04984*, 2017.
- [31] Ruihan Zhao, Kevin Lu, Pieter Abbeel, and Stas Tiomkin. Efficient Empowerment Estimation for Unsupervised Stabilization. *arXiv:2007.07356*, 2021.
- [32] Rein Houthooft, Xi Chen, Yan Duan, John Schulman, Filip De Turck, and Pieter Abbeel. VIME: Variational Information Maximizing Exploration. *Neural Information Processing Systems*, 2017.
- [33] Thomas M Cover and Joy Thomas. *Elements of Information Theory*. John Wiley & Sons, second edition, 2006.
- [34] Alison L Gibbs and Francis Edward Su. On choosing and bounding probability metrics. *International statistical review*, 70(3):419–435, 2002.

- [35] Benjamin Bloem-Reddy and Yee Whye Teh. Probabilistic Symmetries and Invariant Neural Networks. *Journal of Machine Learning Research*, 21(90):1–61, 2020.
- [36] Olav Kallenberg. *Foundations of Modern Probability*, volume 99. Springer, Probability Theory and Stochastic Modelling, 2021.
- [37] Christoph Salge, Cornelius Glackin, and Daniel Polani. Approximation of Empowerment in the Continuous Domain. *Complex Systems*, 16(02n03):1250079, 2013.
- [38] Martin L Puterman. *Markov Decision Processes: Discrete Stochastic Dynamic Programming*. John Wiley & Sons, 2014.
- [39] Suguru Arimoto. An Algorithm for Computing the Capacity of Arbitrary Discrete Memoryless Channels. *IEEE Transactions on Information Theory*, 18(1):14–20, 1972.
- [40] Richard Blahut. Computation of Channel Capacity and Rate-Distortion Functions. *IEEE Transactions on Information Theory*, 18(4):460–473, 1972.
- [41] Jianyi Huang and S.P. Meyn. Characterization and Computation of Optimal Distributions for Channel Coding. *IEEE Transactions on Information Theory*, 51(7):2336–2351, 2005.
- [42] Paul J. Goulart and Yuwen Chen. Clarabel: An Interior-Point Solver for Conic Programs With Quadratic Objectives. *arXiv:2405.12762*, 2024.

A Preliminaries for Proofs

We restate four standard results in forms used in our statements. We cite referenced restatements of these results, which are more general versions of the results below.

The first result gives the standard concavity and convexity properties of mutual information [33, Thm. 2.7.4] [25, Thm. 5.3].

Theorem A.1 (Concavity and convexity of mutual information). *Let P_X be a source distribution and let $P_{Y|X}$ be a channel between standard Borel spaces. For a fixed channel, $I(X; Y)$ is concave in P_X . For a fixed source, $I(X; Y)$ is convex in $P_{Y|X}$.*

In our setting, the source is the skill prior $p(z \mid s_0)$ and the channel maps each skill z to a DSOM $p^\pi(s_+ \mid s_0, z)$. Thus, the MI $I^\pi(Z; S_+ \mid S_0 = s_0)$ is concave in the skill prior for fixed skill occupancies and convex in the occupancies for a fixed skill prior.

The second result expresses the channel capacity in information radius form as the smallest worst-case divergence from the channel outputs to a central distribution [25, Sec. 5.3].

Theorem A.2 (Dual form of channel capacity). *Let $P_{Y|X}$ be a channel between standard Borel spaces $\mathcal{X}$ and $\mathcal{Y}$, and write $P_{Y|X=x}$ for the output distribution at input $x \in \mathcal{X}$. Then*

$$\sup_{P_X \in \Delta(\mathcal{X})} I(X; Y) = \inf_{q(y)} \sup_{x \in \mathcal{X}} D_{\text{KL}}(P_{Y|X=x}(y) \parallel q(y)).$$

If this value is finite, the infimum is attained at a unique center q^ .*

In our setting, the input is the skill, $X = Z$, and the output is the future state, $Y = S_+$, so each output distribution $P_{Y|X=x}$ is a skill occupancy $p^\pi(s_+ \mid s_0, z)$. We utilize this dual form to prove Theorem B.2.

The third result compares KL and χ^2 divergence [25, Eq. (7.31)]; see also Gibbs and Su [34] for the general statement.

Theorem A.3 (KL and χ^2 comparison). *For any distributions $P \ll Q$,*

$$D_{\text{KL}}(P \parallel Q) \leq \log(1 + \chi^2(P \parallel Q)).$$

We apply this bound with

$$P = p^{\pi_{\text{eff}}}(z \mid s_0, s_+) \quad \text{and} \quad Q = p(z \mid s_0).$$

Averaging the bound over s_+ gives the χ^2 control of mutual information that appears in Theorem F.1.

The fourth result shows that a measurable transformation of noise can induce any distribution on a standard Borel space [35, Lem. 5] [36, Thm. 8.17].

Theorem A.4 (Noise outsourcing). *Let μ be a probability distribution on a standard Borel space $\mathcal{Z}$. Let U be the uniform distribution on a standard Borel space $\mathcal{V}$. If $\tilde{Z} \sim U$, then there exists a measurable map $T : \mathcal{V} \rightarrow \mathcal{Z}$ such that*

$$T(\tilde{Z}) \sim \mu.$$

We utilize this statement to prove that a uniform skill prior does not lose generality in Lemma D.1.

B Information Geometry of Empowerment

We prove a general theorem of the information radius result from §. 4.1 that applies to both continuous and discrete MDPs. We include the reduction of the result to the finite-state statement found in §. 4.1.

We start with intuition about the information channel, then explain potential simplifications of this information channel. First, note that skill latents $z \in \mathcal{Z}$ can only influence future state distributions via

a skill-conditioned policy. The information channel $P_{S_+|Z}$ is a function of the skill-conditioned policy $\pi \in \Pi_Z$, which sends each skill to the induced occupancy:

$$Z \xrightarrow{\pi} \mathcal{P}(s_0), \quad z \mapsto p^\pi(s_+ | s_0, z).$$

If this mapping is measurable, we can directly reduce the skills to their induced DSOMs. Specifically, there would exist a pushforward map from the skill random variable Z to a random state occupancy P drawn from a *distribution over $\mathcal{P}(s_0)$* (e.g. a distribution over state occupancy distributions) which, by the data-processing inequality from the Markov chain $Z \rightarrow P_Z \rightarrow S_+$ and the sufficiency of P , would imply

$$I(Z; S_+ | S_0) = I(P_Z; S_+ | S_0).$$

Note, however, that not all policies admit a measurable mapping from skill space to occupancy space. We require the additional assumption that the stochastic policies $\pi(a | s, z)$ are jointly measurable over the state and skill space:

Assumption B.1 (Policies measurable in skill and state space). Let $\mathcal{S}$, $\mathcal{A}$, and $\mathcal{Z}$ denote the state space, action space, and skill spaces respectively, which we take to be standard Borel.

Take the skill-conditioned policy space Π_Z to be the space of policies measurable in the state space $\mathcal{S}$ and skill space $\mathcal{Z}$. That is,

$$(s, z) \mapsto \pi(\cdot | s, z) \in \Delta(\mathcal{A})$$

is a measurable function.

Accordingly, the state occupancy set $\mathcal{P}(s_0)$ is now occupancies reachable by measurable policies.

From this assumption, there exists a measurable mapping from $\mathcal{Z} \xrightarrow{\pi} \mathcal{P}(s_0)$. Thus, we can effectively alias the skill latent Z and the random state occupancy measure P_Z . For the information geometry result, this association allows direct application of Theorem A.2 to the information channel from skills to occupancies:

Theorem B.2 (General information geometry of empowerment). *Fix a start state s_0 . Then, with both sides taking values in $[0, \infty]$,*

$$\mathcal{E}_{\text{pot}}(s_0) = \inf_{q(s_+)} \sup_{p \in \mathcal{P}(s_0)} D_{\text{KL}}(p(s_+) \| q(s_+)).$$

If the empowerment is bounded, the infimum is attainable. Let the skill-conditioned policy $\pi \in \Pi_Z$ and the skill source $p(z | s_0)$ over skills $z \in \mathcal{Z}$ attain $\mathcal{E}_{\text{pot}}(s_0)$. Then, the skill marginal is the unique center:

$$q^* = p^\pi(s_+ | s_0).$$

Moreover, for $p(z | s_0)$ -almost every z ,

$$D_{\text{KL}}(p^\pi(s_+ | s_0, z) \| q^*(s_+)) = \mathcal{E}_{\text{pot}}(s_0),$$

where $p^\pi(s_+ | s_0, z)$ is an extreme point of $\mathcal{P}(s_0)$. Thus, an optimal policy places every skill occupancy at the maximal KL radius, while an optimal source induces the average occupancy q^ .*

We now prove Theorem B.2 with the information-radius identity of Theorem A.2.

Proof. Fix $S_0 = s_0$. Each skill z induces a future state occupancy

$$p^\pi(s_+ | s_0, z) \in \mathcal{P}(s_0),$$

and the conditional distribution of S_+ depends on z only through this occupancy. From the measurability assumption Assumption B.1, we may alias skills that induce the same occupancy. Thus, jointly optimizing the policy and skill source is equivalent to choosing a source over $\mathcal{P}(s_0)$ for the channel

$$p \in \mathcal{P}(s_0), \quad S_+ \sim p.$$

In other words, the source is a distribution *over a set of state-occupancy distributions* that give the channel output S_+ .

Applying Theorem A.2 with the random occupancy $P_Z \in \mathcal{P}(s_0)$ as the input and the future state S_+ as the output gives

$$\mathcal{E}_{\text{pot}}(s_0) = \inf_{q(s_+)} \sup_{p \in \mathcal{P}(s_0)} D_{\text{KL}}(p(s_+) \| q(s_+)).$$

Thus, assuming the potential empowerment is finite, a unique center q^* attains the infimum. Now, suppose that the skill policy and the prior attains the maximal capacity $C = \mathcal{E}_{\text{pot}}(s_0)$. For any distribution q , we have that

$$\mathbb{E}_{p(z|s_0)} [D_{\text{KL}}(p^\pi(s_+ | s_0, z) \| q(s_+))] = I^\pi(Z; S_+ | S_0 = s_0) + D_{\text{KL}}(p^\pi(s_+ | s_0) \| q(s_+))$$

following the Golden Formula (25, Thm. 4.1). Taking $q = q^*$, every reachable occupancy has divergence at most C . Hence,

$$C \geq \mathbb{E}_{p(z|s_0)} [D_{\text{KL}}(p^\pi(s_+ | s_0, z) \| q^*(s_+))] \quad (7)$$

$$= C + D_{\text{KL}}(p^\pi(s_+ | s_0) \| q^*(s_+)) \quad (\text{Golden formula})$$

$$\geq C. \quad (8)$$

Thus, the KL between the marginal state occupancy and q^* is 0, which implies

$$q^* = p^\pi(s_+ | s_0).$$

and that every individual skill divergence is at most C :

$$D_{\text{KL}}(p^\pi(s_+ | s_0, z) \| q^*(s_+)) = C \quad (9)$$

for $p(z | s_0)$ -almost every z . Thus, an optimal source places skill occupancy mass on the boundary of the optimal KL ball, with the KL measured from the central marginal distribution q^* .

Finally, suppose there exists an active skill occupancy that is not an extreme point of $\mathcal{P}(s_0)$. Then, for distinct $p_1, p_2 \in \mathcal{P}(s_0)$ and some $\lambda \in (0, 1)$,

$$p^\pi(s_+ | s_0, z) = \lambda p_1 + (1 - \lambda) p_2.$$

Convexity of the KL divergence in the first argument gives

$$\begin{aligned} D_{\text{KL}}(p^\pi(s_+ | s_0, z) \| q^*(s_+)) &< \lambda D_{\text{KL}}(p_1(s_+) \| q^*(s_+)) + (1 - \lambda) D_{\text{KL}}(p_2(s_+) \| q^*(s_+)) \\ &\leq C, \end{aligned}$$

which contradicts the equality in Eq. (9). Thus, $p(z | s_0)$ -almost every skill induces an extreme point of $\mathcal{P}(s_0)$. $\square$

The reduction to the finite tabular case directly follows:

Theorem 4.1 (Potential empowerment is the KL radius of the reachable polytope, extending Eysenbach et al. [23, Lem. 6.1–6.2]). *Let $\mathcal{P}(s_0) \subset \Delta(\mathcal{S})$ be the reachable state-occupancy polytope of Eq. (3). Jointly maximizing the mutual information over the skill source and the skill-conditioned policy solves the KL 1-center localization problem for $\mathcal{P}(s_0)$:*

$$\mathcal{E}_{\text{pot}}(s_0) = \max_{\pi \in \Pi_{\mathcal{Z}}, p(z|s_0)} I^\pi(Z; S_+ | S_0 = s_0) = \min_{q(s_+) \in \Delta(\mathcal{S})} \max_{p \in \mathcal{P}(s_0)} D_{\text{KL}}(p(s_+) \| q(s_+)). \quad (5)$$

Moreover, every joint optimum $(\pi, p(z | s_0))$ places every skill on the boundary of the optimal KL ball measured from the optimal central occupancy q^ :*

$$D_{\text{KL}}(p^\pi(s_+ | s_0, z) \| q^*(s_+)) = \max_{p \in \mathcal{P}(s_0)} D_{\text{KL}}(p(s_+) \| q^*(s_+)) \quad \text{for every skill } z. \quad (6)$$

In other words, the optimized skills span the maximal information radius of the reachable polytope given starting state s_0 .

Proof of Theorem 4.1. In the finite-state setting, $\mathcal{P}(s_0)$ is a polytope and $\mathcal{E}_{\text{pot}}(s_0) \leq \log |\mathcal{S}|$, so the channel capacity is finite. Theorem B.2 shows that the center distribution is the skill-marginal where active skill occupancies lie on the boundary of the optimal KL ball at extreme points of $\mathcal{P}(s_0)$. The extreme points of a polytope are vertices, which proves the result. $\square$

B.1 Log reachability interpretation of potential empowerment in discrete deterministic settings

For completeness, we include the log reachability interpretation of the action sequence potential empowerment (as originally defined in [1]) in discrete deterministic settings. Salge et al. [3] originally present this result.

We begin by defining the action sequence potential empowerment. Let $A_0^{H-1} = (A_0, \dots, A_{H-1}) \in \mathcal{A}^H$ be an open-loop action sequence over the fixed horizon H . Let S_t denote the random variable of the state after t timesteps. Klyubin et al. [1] define (potential) empowerment as the capacity of the channel from this action sequence to the resulting state:

$$\mathcal{E}_{\text{pot}_H}(s) = \max_{p(a_0^{H-1})} I(A_0^{H-1}; S_H \mid S_0 = s).$$

This expression is equivalent to Definition 3.1, replacing Z with the action sequence A_0^{H-1} and S_+ with S_H .

Lemma B.3 (Discrete deterministic potential empowerment is log reachability, adapted from Salge et al. [3]). *Fix a discrete, deterministic MDP and horizon H . Because dynamics are deterministic, the future state S_H is a deterministic function of S_0 and A_0^{H-1} . Then, the reachable set of states in H timesteps takes the form*

$$\mathcal{R}_H(s) = \{s_+ \in \mathcal{S} : \exists a_0^{H-1} \in \mathcal{A}^H \text{ that reaches } S_H = s_+ \text{ from } S_0 = s\},$$

and the empowerment at s is the log of the size of this set:

$$\mathcal{E}_{\text{pot}_H}(s) = \log |\mathcal{R}_H(s)|.$$

Proof. Because S_H is a deterministic function of A_0^{H-1} given $S_0 = s$, the conditional entropy $H(S_H \mid A_0^{H-1}, S_0 = s)$ is 0. Then,

$$I(A_0^{H-1}; S_H \mid S_0 = s) = H(S_H \mid S_0 = s).$$

Because the support of S_H is in the reachable set $\mathcal{R}_H(s)$, the entropy is at most $\log |\mathcal{R}_H(s)|$. By construction, we can choose one action sequence for each reachable $s_+ \in \mathcal{R}_H(s)$ and place uniform mass $p(a_0^{H-1})$ over the chosen action sequences. Thus, the random variable S_H is uniform on $\mathcal{R}_H(s)$ with entropy $\log |\mathcal{R}_H(s)|$. $\square$

C Empowerment and Bottleneck States

We state and prove the hitting-time and temporal-distance statements from §. 4.2.

C.1 Bottleneck states are high-empowerment states

We first define the repeated rollout hitting time from the DSOM. A single rollout from s_0 samples a skill $z \sim p(z \mid s_0)$ and a geometric termination time $\tau \sim \text{Geom}(1 - \gamma)$. The agent executes skill z until τ , where the rollout terminates at $S_+ \triangleq S_\tau$.

Writing $S_+^{(k)}$ for the termination state of the k th rollout from s_0 , the repeated-rollout hitting time measures the first rollout count that hits s_+ :

$$H_{\text{rst}}^\pi(s_0 \rightarrow s_+) \triangleq \inf \left\{ k \geq 1 : S_+^{(k)} = s_+ \right\},$$

where $H_{\text{rst}}^\pi(s_0 \rightarrow s_+ | z)$ is the corresponding count when every rollout only uses skill z . Then, potential empowerment is exactly the largest expected reduction in log hitting time after committing to a skill:

Lemma 4.2 (Bottleneck states are high potential empowerment states). *In a discrete MDP, let the hitting time random variable $H_{\text{rst}}^{\pi_{\text{eff}}}(s_0 \rightarrow s_+)$ be the number of rollouts from s_0 until a rollout hits s_+ , for skills $Z \sim p(z | s_0)$. Let $H_{\text{rst}}^{\pi_{\text{eff}}}(s_0 \rightarrow s_+ | z)$ be the corresponding hitting time when rolling out z every time. Then,*

$$\mathcal{E}_{\text{pot}}(s_0) = \max_{\pi_{\text{eff}} \in \Pi_Z} \mathbb{E}_{p^{\pi_{\text{eff}}}(z, s_+ | s_0)} [\log \mathbb{E}[H_{\text{rst}}^{\pi_{\text{eff}}}(s_0 \rightarrow s_+)] - \log \mathbb{E}[H_{\text{rst}}^{\pi_{\text{eff}}}(s_0 \rightarrow s_+ | z)]] .$$

Proof. The repeated-rollout hitting times are geometric random variables with success probabilities $p^\pi(s_+ | s_0)$ and $p^\pi(s_+ | s_0, z)$, respectively. Hence,

$$\mathbb{E}[H_{\text{rst}}^\pi(s_0 \rightarrow s_+)] = \frac{1}{p^\pi(s_+ | s_0)}, \quad \mathbb{E}[H_{\text{rst}}^\pi(s_0 \rightarrow s_+ | z)] = \frac{1}{p^\pi(s_+ | s_0, z)} .$$

The MI is the expected log ratio of the skill-conditioned to the marginal DSOM, so

$$\begin{aligned} I^\pi(Z; S_+ | S_0 = s_0) &= \mathbb{E}_{p^\pi(z, s_+ | s_0)} \log \frac{p^\pi(s_+ | s_0, z)}{p^\pi(s_+ | s_0)} \\ &= \mathbb{E}_{p^\pi(z, s_+ | s_0)} [\log \mathbb{E}[H_{\text{rst}}^\pi(s_0 \rightarrow s_+)] - \log \mathbb{E}[H_{\text{rst}}^\pi(s_0 \rightarrow s_+ | z)]] . \end{aligned} \quad (10)$$

This identity holds for any policy. Maximizing both sides over effective empowerment policies π_{eff} gives the result. $\square$

C.2 Temporal distances over contiguous skill rollouts

The above result considers the empowerment objective used throughout the rest of the paper, which measures the controllability of a single skill on the termination state/discounted future observation. We now define a closely related objective over *contiguous* rollouts where the temporal distance interpretation from Myers et al. [6] is exact.

Rather selecting a single skill and rolling out until termination, the agent can chain contiguous skill rollouts into a single trajectory. An agent can begin by selecting a skill z . At some geometric termination time $\tau \sim \text{Geom}(1 - \gamma)$, the agent resamples a random skill and continues forth from the terminated state. This process repeats forever. Intuitively, this rollout scheme is a generalization of standard MDP dynamics, where skill Z now takes the role of the action A , and the β discount factor over units of rollouts controls the fundamental timescale, in addition to the standard γ .

We can mathematically write out the state occupancy measure of such a process. Let the random variable X_k be the state occupied at the k -th geometric termination time. Let the trajectory begin at state $X_0 = s_0$. At the k -th termination, the agent re-draws a skill $Z_k \sim p(z | X_k)$ and continues the trajectory. Then, the state occupancy of the process takes the form

$$M_\beta^\pi(g | s) \triangleq (1 - \beta) \sum_{k=0}^{\infty} \beta^k p^\pi(X_k = g | X_0 = s) .$$

We can correspondingly define the state and skill-conditioned DSOM of the contiguous process as

$$M_\beta^\pi(g | s, z) \triangleq (1 - \beta) \sum_{k=0}^{\infty} \beta^k p^\pi(X_k = g | X_0 = s, Z_0 = z) ,$$

where the agent rolls out skill z until the first geometric time, then resamples skills from the prior $p(z | s_k)$ thereafter. Note that averaging the skill-conditioned DSOM gives the state-conditioned DSOM:

$$\mathbb{E}_{z \sim p(z | s)} [M_\beta^\pi(g | s, z)] = M_\beta^\pi(g | s) . \quad (11)$$

The **effective contiguous empowerment** measures the degree to which the first skill controls the occupancy of the entire chain. Let $Z \sim p(z \mid s_0)$ be the first skill and let $G \sim M_\beta^\pi(g \mid s_0, Z)$ be the state the process occupies at a geometrically sampled termination time. Taking the supremum over the policy and source gives the channel capacity form, or the **potential contiguous empowerment**:

$$\begin{aligned}\mathcal{E}_{\text{eff}}^{\text{contig}}(\pi, s_0) &\triangleq I^\pi(Z; G \mid X_0 = s_0), \\ \mathcal{E}_{\text{pot}}^{\text{contig}}(s_0) &\triangleq \sup_{\pi \in \Pi_Z, p(z \mid s)} \mathcal{E}_{\text{eff}}^{\text{contig}}(\pi, s_0).\end{aligned}$$

An agent is effectively contiguously empowered if their initial skill/generalized action exerts control on their future state for the contiguous process. An agent is potentially contiguously empowered when occupying a state from which immediate skills provide long-term control over the future. We highlight that this modification of empowerment is conceptually similar to the definitions of empowerment considered in the rest of the paper, despite different mathematical forms.

The potential contiguous empowerment admits a temporal distance interpretation, following Myers et al. [6]. The temporal distance captures a notion of the time to traverse between states that generalizes to continuous settings. Let β be the discount factor. Let $M_\beta^\pi(g \mid s)$ denote the DSOM of state g for a process that begins at s . Then, the temporal distance from a state s to a future state g is defined as

$$d_\beta^\pi(s, g) \triangleq -\log \frac{M_\beta^\pi(g \mid s)}{M_\beta^\pi(g \mid g)}$$

where the denominator measures the self-reachability probability for a process that begins at g . Conditioning on the initial skill gives the skill-conditioned temporal distance

$$d_\beta^\pi(s, g \mid z) \triangleq -\log \frac{M_\beta^\pi(g \mid s, z)}{M_\beta^\pi(g \mid g)},$$

again normalized by the self-reaching probability. Let $M_\beta^\pi(z, g \mid s_0) = p(z \mid s_0) M_\beta^\pi(g \mid s_0, z)$ be the joint law of the initial skill Z and future state G . Adding and subtracting the self-reaching term unifies the contiguous empowerment and the temporal distance:

Proposition C.1 (Contiguous empowerment measures temporal distance reduction). *For any skill policy $\pi \in \Pi_Z$ and skill source $p(z \mid s)$,*

$$\mathcal{E}_{\text{eff}}^{\text{contig}}(\pi, s_0) = \mathbb{E}_{M_\beta^\pi(z, g \mid s_0)} [d_\beta^\pi(s_0, g) - d_\beta^\pi(s_0, g \mid z)].$$

Consequently,

$$\mathcal{E}_{\text{pot}}^{\text{contig}}(s_0) = \sup_{\pi \in \Pi_Z, p(z \mid s)} \mathbb{E}_{M_\beta^\pi(z, g \mid s_0)} [d_\beta^\pi(s_0, g) - d_\beta^\pi(s_0, g \mid z)].$$

Thus, the potential contiguous empowerment is the maximum expected reduction in temporal distance to future states when committing to an initial skill Z .

Proof. The result follows directly from definitions. The self-reaching term cancels:

$$d_\beta^\pi(s_0, g) - d_\beta^\pi(s_0, g \mid z) = \log \frac{M_\beta^\pi(g \mid s_0, z)}{M_\beta^\pi(g \mid s_0)}.$$

Taking the expectation over $M_\beta^\pi(z, g \mid s_0)$ gives the equality between the temporal distance reduction and effective contiguous empowerment. Taking the supremum of both sides over the skill-conditioned policy and source gives the potential contiguous empowerment expression. $\square$

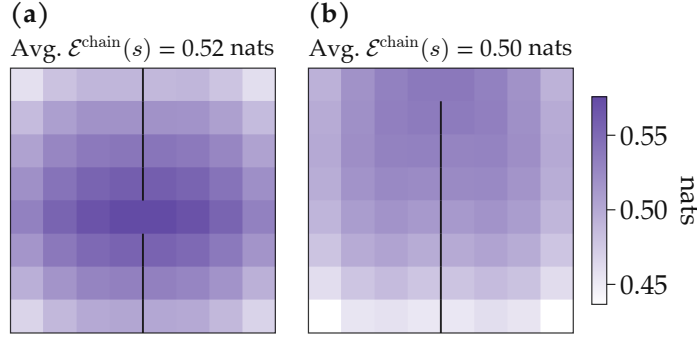


Figure 8: **Empowerment over contiguous rollouts concentrates at bottlenecks** (Proposition C.1). MI of the contiguous process at each start state on the 5×5 grids in Fig. 4, with (a) a center opening and (b) a corner opening. The bottleneck openings have high potential contiguous empowerment.

Unlike the hitting time expression, this form of the contiguous potential empowerment holds in continuous settings. In discrete spaces, the temporal distance reduces to a typical hitting time interpretation [6]. Let

$$H_{\text{contig}}^{\pi}(s \rightarrow g) \triangleq \inf\{k \geq 0 : X_k = g\}$$

be the time to hit g from s , counted in units of contiguous skill rollouts. Let $H_{\text{contig}}^{\pi}(s \rightarrow g | z)$ be the corresponding skill-conditioned hitting time. Following Myers et al. [6], we can rewrite the temporal distance in terms of these hitting times:

$$d_{\beta}^{\pi}(s, g) = -\log \mathbb{E} \left[\beta^{H_{\text{contig}}^{\pi}(s \rightarrow g)} \right], \quad d_{\beta}^{\pi}(s, g | z) = -\log \mathbb{E} \left[\beta^{H_{\text{contig}}^{\pi}(s \rightarrow g | z)} \right] \quad \text{for } g \neq s.$$

Thus, there is a clear hitting time interpretation of the temporal distance object in the discrete limit.

Note on different empowerment expressions. The contiguous empowerment object differs from the empowerment objective used elsewhere in the paper. All other statements in the paper characterize the original potential empowerment; this result is an extension that provides a neat connection to temporal distance as defined in [6], which (unlike hitting time) remains well-defined in continuous settings. Figure 8 shows the MI of the contiguous process from each start state for the Fig. 4 MDPs. The resulting potential contiguous empowerment field closely matches the potential empowerment field considered elsewhere in the paper in Fig. 4. In both experiments, the bottleneck states have high empowerment.

D Skill Selection: Fixed Continuous Uniform Priors are General

Lemma D.1 (A continuous uniform latent can represent any source distribution, discrete or continuous). *Fix a start state s_0 and let the skill alphabet $\mathcal{Z}$ be standard Borel. Consider the mutual information $I^{\pi}(Z; S_+ | S_0 = s_0)$ under a skill-conditioned policy $\pi \in \Pi_{\mathcal{Z}}$, where $Z \sim p(z | s_0)$ is sampled once at s_0 and held fixed for the rollout. Let the reference source U be a continuous distribution on $\mathcal{Z}$, placing zero mass on every single point, e.g. the uniform distribution on $\mathcal{Z}$. Then*

$$\sup_{\pi \in \Pi_{\mathcal{Z}}, p(z|s_0) \in \Delta(\mathcal{Z})} I^{\pi}(Z; S_+ | S_0 = s_0) = \sup_{\tilde{\pi} \in \Pi_{\mathcal{Z}}, \tilde{Z} \sim U} I^{\tilde{\pi}}(\tilde{Z}; S_+ | S_0 = s_0),$$

where the maximization on the right ranges over all skill-conditioned policies.

Proof. Fix any pair (π, p) with $p \in \Delta(\mathcal{Z})$. By the transfer theorem (Theorem A.4; 36), there exists a measurable map $T_{s_0} : \mathcal{Z} \rightarrow \mathcal{Z}$ such that $T_{s_0}(\tilde{Z}) \sim p(z | s_0)$ for $\tilde{Z} \sim U$.¹ Set $Z = T_{s_0}(\tilde{Z})$. Define

$$\tilde{\pi}(a | s, \tilde{z}; s_0) = \pi(a | s, T_{s_0}(\tilde{z}); s_0).$$

¹For $U = \text{Unif}[0, 1]$ this can be the inverse-CDF construction.

Conditioned on $Z = T_{s_0}(\tilde{Z})$, the action kernels of π and $\tilde{\pi}$ agree at every timestep and the environment dynamics are identical. Therefore, the joint law of (Z, S_+) under (π, p) is the same as the joint law of $(T_{s_0}(\tilde{Z}), S_+)$ under $(\tilde{\pi}, U)$. Moreover, $\tilde{\pi}$ only receives information from $\tilde{z}$ through $T_{s_0}(\tilde{z})$, so S_+ is conditionally independent of $\tilde{Z}$ given $T_{s_0}(\tilde{Z})$, and

$$I^{\tilde{\pi}}(\tilde{Z}; S_+ | S_0 = s_0) = I^{\tilde{\pi}}(T_{s_0}(\tilde{Z}); S_+ | S_0 = s_0) = I^{\pi}(Z; S_+ | S_0 = s_0).$$

Taking the supremum over (π, p) shows that the RHS upper bounds the LHS. Conversely, any policy $\tilde{\pi}$ with source $U \in \Delta(\mathcal{Z})$ is within the domain of maximization on the LHS, so we have an equality and are done. $\square$

E Continuous versus Discrete Skill Spaces

Lemma E.1 (Uniform skill priors are not generally optimal with finitely many skills). *With a finite skill alphabet, fixing the skill prior to be uniform can strictly reduce channel capacity.*

Proof. Consider an MDP consisting of a start state s_0 and two terminal states $\{s_1, s_2\}$. Let the skill set be the discrete skill set $\mathcal{Z} = \{0, 1\}$. Let the action space be a continuous action $a \in [0, 1]$. The transition is

$$p(s_1 | s_0, a) = \beta a, \quad p(s_2 | s_0, a) = 1 - \beta a, \quad 0 < \beta < 1.$$

We can fully represent any deterministic, skill-conditioned policy by assigning each skill z to an action $a_z \in [0, 1]$ that completely determines the transition probability $p(s_1 | s_0, z) = \beta a_z$.

First optimize the channel for a fixed source. By convexity of MI in the channel (Theorem A.1; 25, 33), the maximum over $(a_0, a_1) \in [0, 1]^2$ occurs at a vertex. The vertices $(0, 0)$ and $(1, 1)$ transmit no information. The informative vertices are $(0, 1)$ and $(1, 0)$, which are equivalent up to relabeling. Without loss of generality, set $(a_0, a_1) = (0, 1)$, so that $p(s_1 | s_0, z = 0) = 0$ and $p(s_1 | s_0, z = 1) = \beta$.

Let $h(x) = -x \log x - (1 - x) \log(1 - x)$ denote the binary entropy. Under this channel, the output marginal is $p(s_1 | s_0) = p(Z = 1 | s_0) \beta$, and the conditional output entropies are $h(0) = 0$ and $h(\beta)$. The MI is therefore

$$I(Z; S_+ | S_0 = s_0) = h(p(Z = 1 | s_0) \beta) - p(Z = 1 | s_0) h(\beta).$$

Differentiating in the source probability $p(Z = 1 | s_0)$ gives

$$\frac{\partial I(Z; S_+ | S_0 = s_0)}{\partial p(Z = 1 | s_0)} = \beta \log \frac{1 - p(Z = 1 | s_0) \beta}{p(Z = 1 | s_0) \beta} - h(\beta),$$

so the capacity-achieving source is

$$p^*(Z = 1 | s_0) = \frac{1}{\beta(1 + \exp(h(\beta)/\beta))}.$$

For $\beta = 1/2$, this gives $p^*(Z = 1 | s_0) = 2/5$, not $1/2$.

To see that the uniform prior is strictly worse, maximize over channels with $p(Z = 1 | s_0) = 1/2$. The same convexity argument places the optimum at vertex $(a_0, a_1) = (0, 1)$ or $(1, 0)$. For $\beta = 1/2$, the best uniform-prior value is

$$h(1/4) - \frac{1}{2}h(1/2),$$

whereas the best arbitrary source value is

$$h(1/5) - \frac{2}{5}h(1/2).$$

The latter is strictly larger. Thus a uniform prior over two discrete skills cannot generally achieve the channel capacity. $\square$

Comment on VIC [16] and DIAYN [17]. A fixed continuous uniform prior is fully general for discrete and continuous MDPs, while a discrete prior is not. This result has implications for skill learning algorithms. VIC optimizes both a source distribution and channel [16], with options (termed here as “skills”) that may be finite or continuous. However, the authors note that the joint optimization is difficult and practically implement a uniform prior over a discrete skill set. DIAYN similarly fixes the source to a uniform categorical over a discrete skill set [17]. While neither work claims that fixing the prior preserves the objective, our result Lemma E.1 is a concrete counterexample that shows uniform discrete priors lose generality. Finally, Lemma D.1 shows that utilizing a *continuous* uniform skill prior recovers full generality *without the need for source learning*. Combined, our results theoretically suggest that skill learning methods should utilize continuous uniform sources; note that SOTA skill learning methods already utilize such priors [18, 19].

F Empowerment Lower Bounds Tabular Adaptation

We now prove that empowerment lower bounds zero-shot adaptation in the tabular setting under an uninformed reward prior.

Theorem F.1 (Tabular empowerment lower bounds reward adaptation). *Consider the start state s_0 in state space $\mathcal{S}$. Let the skill-marginal DSOM be $p^{\pi_{\text{eff}}}(s_+ | s_0) = \mathbb{E}_{p(z|s_0)}[p^{\pi_{\text{eff}}}(s_+ | s_0, z)]$ of Eq. (2). Fix a reference distribution b on $\mathcal{S}$ with full support (e.g. the uniform baseline $b(s_+) = 1/|\mathcal{S}|$), and sample rewards from the Gaussian*

$$r(s_+) \sim \mathcal{N}\left(0, \frac{1}{b(s_+)}\right).$$

Define the linear adaptation objective

$$J(s_0) \triangleq \mathbb{E}_r \left[\sup_{z \in \mathcal{Z}} \mathbb{E}_{p^{\pi_{\text{eff}}}(s_+ | s_0, z)}[r(s_+)] \right],$$

which calculates the average adapted mean reward. Then, for any skill space $\mathcal{Z}$ and any skill prior $p(z | s_0)$,

$$J(s_0) \geq \frac{I(\mathcal{Z}; S_+ | S_0 = s_0)}{c\sqrt{2\pi}}, \quad c \triangleq \sup_{x \geq 0} \frac{\ln(1+x)}{\sqrt{x}} \approx 0.8.$$

Define the skill offset $u_z \triangleq (p^{\pi_{\text{eff}}}(s_+ | s_0, z) - p^{\pi_{\text{eff}}}(s_+ | s_0))/\sqrt{b}$. The whitened reward $g \triangleq \sqrt{b} r$ is standard normal under the prior, and the marginal’s return $\langle g, p^{\pi_{\text{eff}}}(s_+ | s_0)/\sqrt{b} \rangle$ is mean-zero and z -independent, so

$$J(s_0) = \mathbb{E}_g \sup_{z \in \mathcal{Z}} \langle g, u_z \rangle, \tag{12}$$

is the Gaussian width of $\{u_z\}_{z \in \mathcal{Z}}$.

Before presenting the proof, we discuss the chosen reward prior and motivate the variance scaling. Under the $1/b$ scaling, splitting a state into substates leaves the distribution of the returns unchanged.

Example F.2 (Splitting a two-state MDP into a four-state MDP Fig. 9). Let $\mathcal{S} = \{1, 2\}$ with uniform reference $b(1) = b(2) = \frac{1}{2}$. The prior samples $r(1), r(2) \sim \mathcal{N}(0, 2)$ independently (Fig. 9). For a skill z with DSOM $p^{\pi_{\text{eff}}}(s_+ | s_0, z) = (p_1, p_2)$, the mean reward is a centered Gaussian:

$$\mathbb{E}_{p^{\pi_{\text{eff}}}(s_+ | s_0, z)}[r(s_+)] = p_1 r(1) + p_2 r(2) \sim \mathcal{N}(0, 2p_1^2 + 2p_2^2).$$

Splitting each state into two substates gives $\mathcal{S}' = \{1a, 1b, 2a, 2b\}$ where the refined prior samples each coordinate reward $r' \sim \mathcal{N}(0, 4)$. Furthermore, split every skill’s occupancy in the same proportions, so the occupancy of z becomes $(\frac{p_1}{2}, \frac{p_1}{2}, \frac{p_2}{2}, \frac{p_2}{2})$. The mean reward on the fine-grained space is

$$\frac{p_1}{2}(r'(1a) + r'(1b)) + \frac{p_2}{2}(r'(2a) + r'(2b)) \sim \mathcal{N}\left(0, \frac{p_1^2}{4} \cdot 8 + \frac{p_2^2}{4} \cdot 8\right) = \mathcal{N}(0, 2p_1^2 + 2p_2^2),$$

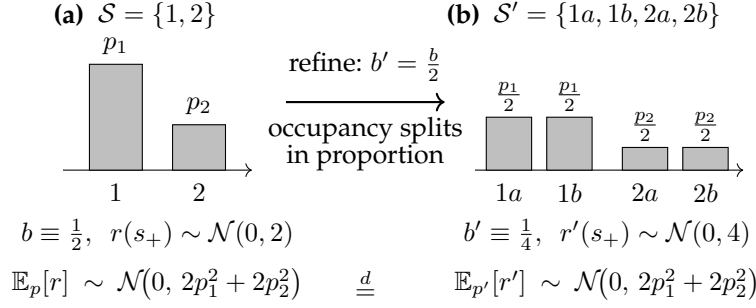


Figure 9: **State refinement leaves return distribution unchanged (Example F.2).** Bars show the occupancy $p^{\pi_{\text{eff}}}(s_+ | s_0, z) = (p_1, p_2)$ of a skill over the coarse states (a) and its occupancy over the split states (b). Each panel lists the reference mass and per-state reward prior $r(s_+) \sim \mathcal{N}(0, 1/b(s_+))$. Splitting states halves the reference mass and doubles the reward variance. The return has the same law $\mathcal{N}(0, 2p_1^2 + 2p_2^2)$ in both MDPs.

since each sum $r'(\cdot a) + r'(\cdot b)$ has variance 8. Thus, the distribution of the mean reward is the same before and after splitting the state.

In this example, the $1/b$ variance scaling is key. Under a unit-variance prior $r(s_+) \sim \mathcal{N}(0, 1)$, the same state splitting shrinks the return variance from $p_1^2 + p_2^2$ to $\frac{1}{2}(p_1^2 + p_2^2)$. Thus, the constraint that the return distribution is scale invariant implies the above scaling of the reward prior variance.

Now that we have built intuition for the choice of reward prior, we present the proof:

Proof. By Eq. (12), it suffices to lower bound the Gaussian width. The proof consists of two inequalities:

$$J(s_0) \stackrel{\text{step 1}}{\geq} \frac{1}{\sqrt{2\pi}} \left(\mathbb{E}_{p(z|s_0)} \|u_z\|^2 \right)^{1/2} \stackrel{\text{step 2}}{\geq} \frac{I(Z; S_+ | S_0 = s_0)}{c\sqrt{2\pi}}. \quad (13)$$

Fix the reward r and a skill $z \in \mathcal{Z}$. For the first inequality (step 1), we observe that the best average return obtainable by a skill dominates the marginal skill return:

$$\sup_{z' \in \mathcal{Z}} \mathbb{E}_{p^{\pi_{\text{eff}}}(s_+ | s_0, z')} [r(s_+)] \geq \max \left(\mathbb{E}_{p^{\pi_{\text{eff}}}(s_+ | s_0, z)} [r(s_+)], \mathbb{E}_{p^{\pi_{\text{eff}}}(s_+ | s_0)} [r(s_+)] \right).$$

Take the expectation over rewards on both sides using the identity $\max(x, y) = \frac{1}{2}(x + y) + \frac{1}{2}|x - y|$. Both returns are linear in r , so the $\frac{1}{2}(x + y)$ term vanishes and

$$J(s_0) \geq \frac{1}{2} \mathbb{E}_r \left| \mathbb{E}_{p^{\pi_{\text{eff}}}(s_+ | s_0, z)} [r(s_+)] - \mathbb{E}_{p^{\pi_{\text{eff}}}(s_+ | s_0)} [r(s_+)] \right|.$$

The gap in returns is an inner product of the whitened quantities:

$$\mathbb{E}_{p^{\pi_{\text{eff}}}(s_+ | s_0, z)} [r(s_+)] - \mathbb{E}_{p^{\pi_{\text{eff}}}(s_+ | s_0)} [r(s_+)] = \langle g, u_z \rangle.$$

Because the coordinates of g are i.i.d. standard Gaussians, we know that the gap in returns is a Gaussian with variance $\|u_z\|^2$. Hence, for every $z \in \mathcal{Z}$,

$$\begin{aligned} J(s_0) &\geq \frac{1}{2} \mathbb{E}_r |\langle g, u_z \rangle| \\ &= \frac{1}{\sqrt{2\pi}} \|u_z\|. \end{aligned}$$

Averaging the square of this bound over $z \sim p(z | s_0)$ (LHS is independent of the skill z) gives step 1 of Eq. (13). For step 2, express the MI as the expected KL divergence between the skill prior and the skill posterior after observing s_+ :

$$I(Z; S_+ | S_0 = s_0) = \sum_{s_+} p^{\pi_{\text{eff}}}(s_+ | s_0) D_{\text{KL}}(p^{\pi_{\text{eff}}}(z | s_0, s_+) \| p(z | s_0)).$$

Applying Bayes' rule, the χ^2 divergence takes the form of a mean squared deviation of the skill DSOMs from the marginal:

$$\begin{aligned}\chi^2(p^{\pi_{\text{eff}}}(z | s_0, s_+) || p(z | s_0)) &= \mathbb{E}_{p(z|s_0)} \left[\left(\frac{p^{\pi_{\text{eff}}}(s_+ | s_0, z)}{p^{\pi_{\text{eff}}}(s_+ | s_0)} - 1 \right)^2 \right] \\ &= \frac{\mathbb{E}_{p(z|s_0)} [(p^{\pi_{\text{eff}}}(s_+ | s_0, z) - p^{\pi_{\text{eff}}}(s_+ | s_0))^2]}{p^{\pi_{\text{eff}}}(s_+ | s_0)^2}.\end{aligned}$$

Then, through χ^2 , we can upper bound the MI with the norm of the occupancy gap vector:

$$\begin{aligned}I(Z; S_+ | S_0 = s_0) &\leq c \sum_{s_+} p^{\pi_{\text{eff}}}(s_+ | s_0) \sqrt{\chi^2(p^{\pi_{\text{eff}}}(z | s_0, s_+) || p(z | s_0))} \quad (\text{Theorem A.3}) \\ &= c \sum_{s_+} \sqrt{\mathbb{E}_{p(z|s_0)} [(p^{\pi_{\text{eff}}}(s_+ | s_0, z) - p^{\pi_{\text{eff}}}(s_+ | s_0))^2]} \\ &= c \sum_{s_+} \sqrt{b(s_+)} \sqrt{\frac{\mathbb{E}_{p(z|s_0)} [(p^{\pi_{\text{eff}}}(s_+ | s_0, z) - p^{\pi_{\text{eff}}}(s_+ | s_0))^2]}{b(s_+)}} \\ &\leq c \left(\sum_{s_+} \frac{\mathbb{E}_{p(z|s_0)} [(p^{\pi_{\text{eff}}}(s_+ | s_0, z) - p^{\pi_{\text{eff}}}(s_+ | s_0))^2]}{b(s_+)} \right)^{1/2} \quad (\text{Cauchy-Schwarz}) \\ &= c \left(\mathbb{E}_{p(z|s_0)} \|u_z\|^2 \right)^{1/2}.\end{aligned}$$

The first line applies Theorem A.3 with the bound $\ln(1+x) \leq c\sqrt{x}$ to each divergence. The second line expands χ^2 and cancels the marginal weight. The third and fourth lines apply Cauchy-Schwarz. The last equality is the definition of u_z , proving step 2 of Eq. (13) which finishes the proof. $\square$

We provide some didactic tabular experiments that demonstrate the behavior and scaling of the bound over different MDPs. In Fig. 10, we demonstrate the relationship between the degree of commitment within an MDP – the controllability of the MDP – and the bound for a fixed set of states. As controllability of the MDP increases, the bound tightens. In Fig. 11, we demonstrate the relationship between the number of blocks in an MDP (e.g. number of relative states) and the bound under fixed commitment. As the number of blocks grows, the bound loosens.

G When Information Geometry is *Insufficient* for Adaptation

The adaptation result above is in the tabular setting. For the following results, we also consider MDPs whose state space $\mathcal{S}$ is a metric space with arbitrary metric d . Importantly, these spaces may be continuous.

Proposition G.1 (MDPs can admit arbitrarily many distinct skills). *Fix a discount $\gamma \in (0, 1)$ and a skill space $\mathcal{Z} = [K]$ with the uniform prior $Z \sim \text{Unif}([K])$. There exists an MDP with a start state $s \in \mathcal{S}$ and K skills that attains the potential empowerment*

$$\mathcal{E}_{\text{pot}}(s) = \max_{\pi_{\text{eff}}} I^{\pi_{\text{eff}}}(Z; S_+ | S_0 = s) = \gamma \log K.$$

If $\mathcal{S}$ is continuous, there exists an MDP where a continuum of these skills gives unbounded potential empowerment $\mathcal{E}_{\text{pot}}(s) = +\infty$.

Proof. Discrete case. We construct MDPs that admit $\mathcal{E}_{\text{pot}}(s) \geq \gamma \log K$ for all K .

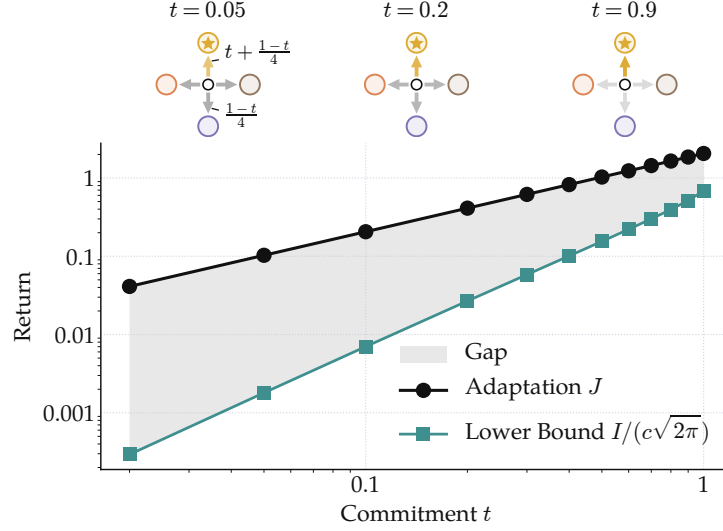


Figure 10: **MI bound as a function of skill commitment** (Theorem F.1). Consider the MDP consisting of 5 states with a central starting state, where any left/right/up/down action sends probability mass to absorbing left/right/up/down states. Actions left/right/up/down have a level of commitment t , which measures the additional mass above uniform assigned to the corresponding state. When commitment is small, probability is almost spread equally over all four states regardless of action so there is little control. In this regime, the adaptation J dominates the lower bound. As the commitment increases, actions/skills commit to states and the bound tightens.

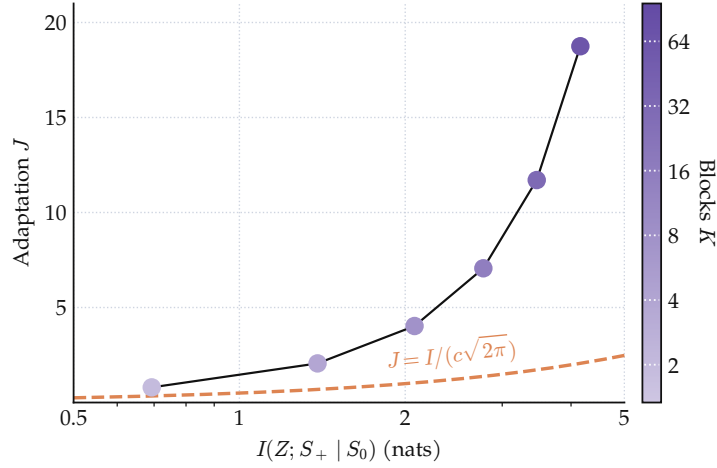


Figure 11: **MI bound as a function of distinct state regions** (Theorem F.1). Consider the MDP consisting of 128 absorbing states split into K distinct blocks. Each of the K actions sends probability uniformly over the K th block. Thus, K counts the maximum possible number of total skills. The scaled MI (orange) correctly lower bounds the adaptation J (purple). The bound weakens as the number of blocks in the MDP grows.

Consider an MDP that admits a large number of distinguishable occupancies. Then, we can assign each skill a distinct occupancy at some target future state. In this limit, the skill becomes readily identifiable from the future state.

For finite K , consider a discretized state space $\mathcal{S}$ with states $s, s_1, \dots, s_K \in \mathcal{S}$ where all states $s_1, \dots, s_K$ are absorbing. Let skill z immediately transition from start state s to another target state. Then, the induced DSOM is

$$p^{\pi^{\text{eff}}}(s_+ | s, z) = (1 - \gamma) \delta_s + \gamma \delta_{s_z},$$

where δ is the Kronecker delta. Here, the skills share the start state s with probability mass $1 - \gamma$ and are otherwise disjoint over $\mathcal{S} \setminus \{s\}$. Observing $S_+ \neq s$ exactly determines the skill while observing $S_+ = s$ gives no information. Thus, the MI is

$$I^{\pi_{\text{eff}}}(Z; S_+ | S_0 = s) = \gamma H(Z) = \gamma \log K,$$

which obtains the potential empowerment.

Continuous case. We similarly construct MDPs that admit unbounded $\mathcal{E}_{\text{pot}}(s)$. When $\mathcal{S}$ is continuous, we can index skills by a continuous parameter rather than a finite set and map $z \in [0, 1]$ to $\mathcal{S} \setminus \{s\}$. The induced DSOM is

$$p(s_+ | s, z) = (1 - \gamma) \delta_s + \gamma \delta_{\phi(z)}$$

where δ is now the Dirac delta. As before, any final state that is not the start state identifies the skill. If we cut $[0, 1]$ into n equal segments, the future state S_+ reveals at least $\gamma \log n$ bits about the skill. Since this construction holds for all n , we have that

$$I(Z; S_+ | S_0 = s) \geq \sup_n \gamma \log n = +\infty.$$

Thus, we have constructed an MDP where the MI is unbounded for every $\gamma \in (0, 1)$ for continuous state spaces. $\square$

We can utilize this proposition to prove that potentially infinite bits can fit within a tiny metric ball:

Proposition G.2 (Many bits can fit in a tiny metric ball). *Fix a start state s and a ball $B(\bar{s}, \epsilon)$ of radius ϵ with infinitely many states, centered at an arbitrary state $\bar{s} \in \mathcal{S}$. For every K , there exists an MDP with K skills as in Proposition G.1 whose target states s_z all lie inside the ball, so*

$$I(Z; S_+ | S_0 = s) = \gamma \log K,$$

yet every bounded 1-Lipschitz reward r gives all skills nearly the same return:

$$\sup_{z, z'} |\mathbb{E}_{p(s_+ | s, z)}[r(S_+)] - \mathbb{E}_{p(s_+ | s, z')}[r(S_+)]| \leq 2\gamma\epsilon \leq 2\epsilon.$$

With a continuum of skills inside the ball, the MI can be unbounded while the upper bound on the return difference between skills is unchanged.

Proof. We can apply Proposition G.1 to the ball $B(\bar{s}, \epsilon)$ and construct valid MDPs with the above MIs. For any 1-Lipschitz reward, the skill returns of $z \neq z'$ in these MDPs differ by

$$\begin{aligned} |\mathbb{E}_{p(s_+ | s, z)}[r(S_+)] - \mathbb{E}_{p(s_+ | s, z')}[r(S_+)]| &= \gamma |r(s_z) - r(s_{z'})| \\ &\leq \gamma (d(s_z, \bar{s}) + d(\bar{s}, s_{z'})) \\ &\leq 2\gamma\epsilon \end{aligned}$$

which proves the result. $\square$

For any 1-Lipschitz reward, it is always possible for a set of skills to have high MI (potentially unbounded) while providing no adaptation advantage for continuous rewards beyond a single policy. Thus, we should *not* extrapolate the tabular adaptation bound (Theorem F.1) to continuous spaces with smooth rewards.

We show an example of this collapse in Fig. 6. The reachable future state occupancies pack closer together in continuous space from left to right. In the limit, the difference in values between skills shrinks to zero while the MI diverges, so the discrete adaptation bound becomes trivial. We note that prior works explicitly point out and propose solutions to the problem of diverging channel capacities in continuous spaces [30, 37], though the works do not formally connect empowerment and adaptation.

H Anisotropic Reward Priors

This section relaxes the reward isotropy assumption in the adaptation result [Theorem F.1](#). Unfortunately, using an anisotropic reward prior with covariance Σ scales the lower bound by $\sqrt{\lambda_{\min}(\Sigma)}$ ([Proposition H.1](#)). In the limit, skills with arbitrarily large mutual information can exhibit no adaptation even with fixed reward power constraints (e.g. fixed $\text{Tr } \Sigma$; see [Proposition H.2](#)).

We keep the discrete setting of [Theorem F.1](#) and only change the reward prior. We sample rewards with $\sqrt{b}r \sim \mathcal{N}(0, \Sigma)$, so the law of the uniform baseline-whitened reward g in [Eq. \(12\)](#) becomes $\mathcal{N}(0, \Sigma)$. An arbitrary anisotropic prior may concentrate variance on directions that the empowerment-maximizing skills choose to ignore. The marginal's return remains mean zero for any Σ , so the adaptation objective takes the form

$$J^\Sigma(s_0) \triangleq \mathbb{E}_{g \sim \mathcal{N}(0, \Sigma)} \left[\sup_{z \in \mathcal{Z}} \langle g, u_z \rangle \right].$$

Proposition H.1 (Anisotropic reward prior scales bound). *In the setting of [Theorem F.1](#), sample rewards with $\sqrt{b}r \sim \mathcal{N}(0, \Sigma)$ instead. Then, the MI bound rescales by $\sqrt{\lambda_{\min}(\Sigma)}$:*

$$J^\Sigma(s_0) \geq \sqrt{\lambda_{\min}(\Sigma)} \frac{I(Z; S_+ | S_0 = s_0)}{c\sqrt{2\pi}}.$$

The isotropic prior $\Sigma = \text{Id}$ recovers [Theorem F.1](#) exactly.

Proof. The proof of [Theorem F.1](#) uses the reward prior to express a lower bound on $J(s_0)$ in terms of the norm of the occupancy gap vector. Under the $\mathcal{N}(0, \Sigma)$ prior, the occupancy gap vector variance scales by the anisotropic covariance. Then, the result from [Theorem F.1](#) suggests the following modified bound:

$$J^\Sigma(s_0) \stackrel{\text{step 1}}{\geq} \frac{1}{\sqrt{2\pi}} \left(\mathbb{E}_{p(z|s_0)} \left\| \Sigma^{1/2} u_z \right\|^2 \right)^{1/2} \stackrel{\text{step 2}}{\geq} \sqrt{\lambda_{\min}(\Sigma)} \frac{I(Z; S_+ | S_0 = s_0)}{c\sqrt{2\pi}}. \quad (14)$$

Now, we carefully derive this bound. The reduction of J to the gap in returns in [Theorem F.1](#) only depends on a centered reward. Thus, as before, we have that

$$J^\Sigma(s_0) \geq \frac{1}{2} \mathbb{E}_g |\langle g, u_z \rangle|.$$

For $g \sim \mathcal{N}(0, \Sigma)$, the return gap $\langle g, u_z \rangle$ is Gaussian with variance $\|\Sigma^{1/2} u_z\|^2$. Hence, for every $z \in \mathcal{Z}$,

$$J^\Sigma(s_0) \geq \frac{1}{\sqrt{2\pi}} \left\| \Sigma^{1/2} u_z \right\|.$$

Averaging the square of this bound over $z \sim p(z | s_0)$ (the left side does not depend on z) gives step 1. We can lower bound the norm:

$$\|\Sigma^{1/2} u_z\|^2 = \langle u_z, \Sigma u_z \rangle \geq \lambda_{\min}(\Sigma) \|u_z\|^2.$$

Combining with the step 2 bound (does not change with the reward prior) gives the desired result. $\square$

Indeed, we can construct an example where a state and set of policies with high potential empowerment exhibit zero adaption to anisotropic rewards:

Proposition H.2 (Large MI with zero anisotropic adaptation). *Fix a power budget $T > 0$. For every K , there exists a discrete MDP with skills achieving $I(Z; S_+ | S_0 = s_0) = \gamma \log K$ and a reward prior $\sqrt{b}r \sim \mathcal{N}(0, \Sigma)$ with $\text{Tr } \Sigma = T$ such that $J^\Sigma(s_0) = 0$.*

Proof. The discrete construction in Proposition G.1 attains $I(Z; S_+ \mid S_0 = s_0) = \gamma \log K$ with K occupancy vector offsets u_z that span a subspace of dimension at most $K - 1$ of the $(K+1)$ -dimensional whitened reward space.

Choose a unit vector v orthogonal to every u_z and set the reward variance $\Sigma = T v v^\top$. This choice places all variance of the reward prior in the space orthogonal to the span of the occupancy gap vectors. By the proof of Proposition H.1, each return gap $\langle g, u_z \rangle$ is Gaussian with variance $\|\Sigma^{1/2} u_z\|^2 = 0$, so $J^\Sigma(s_0) = 0$ while the MI is unchanged. $\square$

Thus, while the MI alone can provide nontrivial lower bounds on adaptation in tabular settings for an isotropic reward prior, the same no longer holds true for anisotropic reward priors. We provide intuition for this theoretical result in Fig. 7. Here, the degrees of freedom with maximal controllability are different from the most rewarding degrees of freedom. An empowerment-maximizing agent would learn skills with high control over observations (left/right movement) and ignore other learnable skills that control the rewarding degrees of freedom (up/down movement).

I Reward Adaptation for General Priors

We can strengthen the result in Theorem F.1 to hold for a more general class of Gaussian priors. The key is that the covariance of the reward prior across states is an inner product in some Hilbert space. High empowerment skills imply a high degree of (KL) separation from the centroid of the skill DSOMs (the marginal state occupancy). This implies a certain degree of Euclidean separation from the centroid, and thus a certain maximal pairwise Euclidean separation between skill DSOMs.

Theorem I.1 (Empowerment lower bound on adaptation for general centered GP reward prior).

J Reconciling Reward and Information Geometry

Reconciling reward geometry with information geometry requires adjusting either reward geometry and/or information geometry:

1. reformulate the value maximization framework and/or algorithms in terms of bits alongside information-theoretic exploration objectives like empowerment, with principled reasoning for why values should take the form of bits (cite the max likelihood RL paper), or
2. reformulate empowerment in value space (e.g. Gaussian width geometry as suggested from the above results) to tightly link empowerment and adaptation.

K Tabular Experiment Details

K.1 Tabular 5×5 grid experiments

Table 1 details the 5×5 grid MDP parameters used for Fig. 2.

Skill-learning via convex optimization. In the state centrality experiments, we used a convex solver to enumerate vertices of the reachable discounted state occupancy measure (DSOM) polytope for all possible starting states. An MDP with $|S|$ states and $|A|$ actions can have up to $|A|^{|S|}$ vertices. Thus, enumerating all vertices is challenging [23].

However, we can use the fact that reward-maximizing DSOM always sit at the vertices of the convex polytope to conduct vertex search. Specifically, we (1) sample a random reward vector $\theta \in \Delta(S)$ in the state simplex and (2) use a convex solver to maximize DSOM alignment with the reward vector subject to probability flow constraints [38]. Since a linear objective over the reachable state polytope is maximized at an extreme point, and a generic random objective exposes a vertex almost surely, this

procedure retrieves valid vertices of the polytope conditioned on specific starting states. This procedure generally does not retrieve the complete set of candidate vertices over which to specify skills, but it provides a good approximation.

Policies that optimize

$$\max_{\pi(a|s,z;s_0), p(z|s_0)} I(Z; S_+ | S_0 = s_0),$$

seek out the vertices of the polytope (see Theorem 4.1). Thus, we fix the approximate feasible set of optimal policies to the set of found vertices and break symmetries among policies with equivalent vertex DSOMs by selecting the policy with minimal entropy. This procedure constructs a one-to-one mapping between policies and vertex DSOMs and fixes the encoder in the information channel. Then, getting the empowerment-maximizing policy with known dynamics is simple: it is well known that for a fixed channel encoder $p^\pi(s_+ | s_0, z)$, the MI is concave in the source distribution $p(z | s_0)$ and convex in the channel (Theorem A.1; 33). Thus, finding the optimal source distribution $p(z | s_0)$ is reduced to a convex optimization problem solvable with standard convex optimization packages (see the Blahut–Arimoto algorithm; 39, 40).

An alternative to sampling random reward vectors to search for policies is to utilize an iterative form of the B-A algorithm that alternates source optimization with a more principled vertex search. We additionally implement a version of Huang and Meyn [41] that uses cutting plane linearization to find occupancy vertices that increase the MI. While this method is more involved, the takeaways of the tabular experiments are the same whether utilizing the random reward vertex search or a more principled method for capacity maximization and vertex search.

The results in Fig. 2 show that the latent z -conditioned policies that start at $s_0 = (2, 2)$ cover all states within the MDP.

Learning empowerment-maximizing policies. The above procedure optimizes the channel capacity over the sampled vertex set, yielding a lower bound on

$$\mathcal{E}_{\text{pot}}(s_0) = \max_{\pi(a|s,z;s_0), p(z|s_0)} I(Z; S_+ | S_0 = s_0)$$

that tightens as the candidate vertex set grows. We use standard Q-learning methods to obtain the greedy policy that maximizes

$$Q^{\pi_{\text{pot}}}(s, a) = \mathbb{E} \left[\sum_{t=0}^{\infty} \gamma^t \mathcal{E}_{\text{pot}}(s_t) \middle| s_0 = s, a_0 = a \right],$$

which we refer to as the potential empowerment Q -value. We anneal the epsilon greedy action rate from $1.0 \rightarrow 0.01$ from the start to the end of Q-learning. The optimal policy shown in Fig. 2(c) moves to the center of the room, maximizing reachability to all other states in the MDP.

K.2 Bottleneck grid experiments

Figure 4 uses the same vertex search, source-distribution optimization, and Q-learning procedure as Appendix K.1. The only difference is that in this GridWorld, the map has impenetrable walls within the state space.

This MDP demonstrates the bottleneck effect discussed in §. 4.2. In the open grid, the center state has maximal potential empowerment. With the added walls, the bottleneck openings at the gaps in the walls have high empowerment. At these states, the agent’s choice in skill substantially changes its temporal distance to future states.

K.3 Key-finding experiments

Table 2 gives the key-finding experiment hyperparameters.

Table 1: Hyperparameters for the 5×5 gridworld experiments (Figs. 2 and 4)

Parameters	Value
Environment size ($n \times m$)	5×5
Environment actions	{Left, Right, Up, Down, Stay}
Periodic boundary conditions	No
$P(\text{random action})$	0.0
Gamma for DSOM	0.9
# uniformly sampled vertex directions per s_0	500
Convex solver	Clarabel [42]
# max iterations	50_000
Error tolerance	$1e-6$
Vertex seed	0
Epsilon schedule	Linear interpolation from $1 \rightarrow 0.01$
Gamma for potential policy	0.9
# Q-learning iterations	2×10^5

We use a numerical experimental setup identical to that of the 5×5 room with walls. In this MDP, the agent can pick up and drop off a key at the square $(0, 2)$. Picking up this key allows access to the blue-shaded regions of the MDP; without the key, the agent cannot enter these spaces in the grid.

Fig. 12 shows the resulting empowerment-maximizing policy that ascends the empowerment landscape. Starting at state $s_0 = (4, 2)$ without a key, the agent learns to navigate to the key, pick it up, and navigate to the center square $(2, 2)$. This experiment shows that states with maximal empowerment capture an information-geometric notion of centrality—rather than simply navigating to and staying at the *physically* center state, as in the 5×5 grid, the agent recognizes that the most central state is the one where it has obtained the key.

 Table 2: Hyperparameters for the 5×5 gridworld with walls and a key

Parameters	Value
Environment size ($n \times m$)	5×5
Environment actions	{Left, Right, Up, Down, Stay, Pick up key at $(0, 2)$, Drop key at $(0, 2)$ }
Periodic boundary conditions	No
$P(\text{random action})$	0.0
Gamma for skill-learning	0.99
# uniformly sampled vertex directions per s_0	1000
Convex solver	Clarabel [42]
# max iterations	50_000
Error tolerance	$1e-6$
Vertex seed	0
Epsilon schedule	Linear interpolation from $1 \rightarrow 0.01$
Gamma for potential policy	0.99
# Q-learning iterations	2×10^5

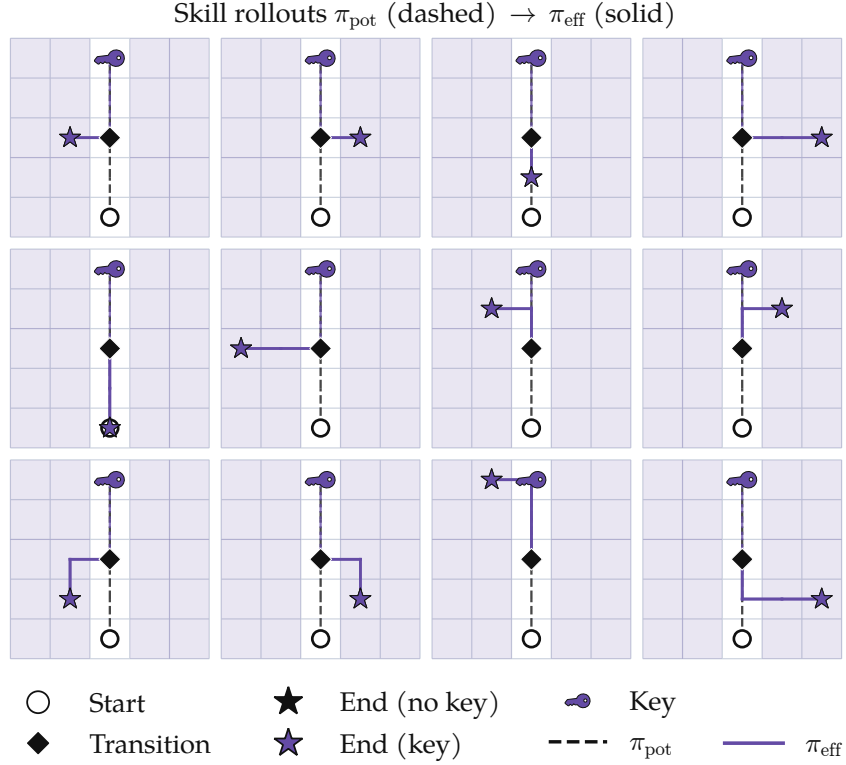


Figure 12: **High empowerment finds central states, where skills are diverse from the central states** (Theorem 4.1). Potential policy navigates to central states. Learned skills spread out from the central state and measure the degree of centrality of the state (in information geometry). Dotted line denotes trajectory while rolling out the potential empowerment maximization policy for $\frac{1}{1-\gamma}$ steps. Diamond is the handoff state s_0 , where $z \sim p(z \mid s_0)$ is randomly sampled and the skill policy $\pi(a \mid s, z; s_0)$ is rolled out.

K.4 Adaptation geometry examples

In each example MDP of Fig. 5 and Fig. 7, every skill transitions in one step from the starting state to an absorbing target state. Thus, a skill’s DSOM, normalized over the targets, is the transition distribution. Table 3 lists the resulting occupancy distributions.

In Example X (Fig. 5(a)), the $N_T = 36$ targets form two halves. We design the MDP such that each of two skills can place mass $a = 0.65$ uniformly on its own half; the MDP spreads the remaining $1 - a$ probability mass uniformly over all N_T targets. In Example Y (Fig. 5(d)), all three skills reach three of the N_T targets and no others. Each skill primarily navigates to one of three targets, placing mass $\frac{1}{2}$ on the target. The MDP dynamics forces $\frac{1}{4}$ probability mass on each of the other two targets.

In the four-block MDP (Fig. 7(a)), the left/right block pair carries total mass 0.45 and the up/down pair carries 0.55. We split the probability mass evenly within each pair, so each block has mass m , half its pair’s total. A skill with relative commitment t scales the probability mass to its own block by a factor of $1 + t$ and the opposing block by $1 - t$, leaving the other pair’s mass unchanged. Averaging over p^* returns every block mass to m , so a committed skill differs from the capacity-achieving mixture q^* only on its own pair and

$$D_{\text{KL}}(p^{\pi_{\text{eff}}}(s_+ \mid s_0, z) \parallel q^*(s_+)) = m[(1+t)\log(1+t) + (1-t)\log(1-t)].$$

Table 3: Skill occupancy distributions of the example MDPs. (a) Example X with total mass on each half of the targets. (b) Example Y with mass on the three reachable targets. (c) the four-block MDP with mass on blocks A (left), B (right), C (up), D (down).

(a) Figure 5(a) MDP			(b) Figure 5(d) MDP				(c) Figure 7(a) MDP				
	Half 1	Half 2		A	B	C		A	B	C	D
$z = 1$	0.825	0.175	$z = 1$	0.50	0.25	0.25	$z = \text{left}$	0.419	0.031	0.275	0.275
$z = 2$	0.175	0.825	$z = 2$	0.25	0.50	0.25	$z = \text{right}$	0.031	0.419	0.275	0.275
			$z = 3$	0.25	0.25	0.50	$z = \text{up}$	0.225	0.225	0.349	0.201
							$z = \text{down}$	0.225	0.225	0.201	0.349

We fix the divergence of each pair to 0.20 nats for the left/right pair and 0.02 nats for the up/down pair to illustrate the empowerment discrepancy, and solve the closed form for the corresponding commitment t . The capacity-achieving source then uses the left/right skills alone, so $\mathcal{E}(s) = 0.20$ nats.

Figure 5(b, e) plot the integrand of Eq. (12) with the uniform baseline $b = 1/N_T$ over the targets. The overlap between Example Y’s skills leads to low channel capacity. However, the spread over the target states gives a larger Gaussian width. The orderings of the adaptation and the obtained MI therefore flip: X attains the higher mutual information and Y the larger width.

In Fig. 7(b), the reward prior sets $\sqrt{b}r \sim \mathcal{N}(0, \Sigma)$ with $\Sigma^{1/2} = \text{diag}(0.55, 2.40)$ in the panel’s coordinates (g_1, g_2) , and equal to the identity on the directions orthogonal to the plane (which contribute zero to every return). We change the reward prior but leave the MDP structure fixed, so the learned skills generally do not align with the sampled rewards (e.g. reward contour avoids regions of the yellow and orange skills).

In the reward panels (Fig. 5(b, e) and Fig. 7(b)), the plane is the span of the offsets u_z of Eq. (12), and the axes (g_1, g_2) give the coordinates of g in an orthonormal basis of this span (under which the isotropic prior of Eq. (12) is again standard normal). Components of g orthogonal to every u_z contribute zero to each $\langle g, u_z \rangle$, so the plots drop those directions.

At most two offsets are linearly independent in each example. The occupancy offsets in Example X’s are negatives of each other since probability sums to 1, Example Y’s three offsets satisfy $u_1 + u_2 + u_3 = 0$, and the four-block offsets satisfy $u_{\text{left}} = -u_{\text{right}}$ and $u_{\text{up}} = -u_{\text{down}}$. Thus, two principle components describe the entire return gap, so we can draw a 2D diagram to describe the return field and information geometry Fig. 5(e); the insets in Fig. 5(f) visualize the principle components. Example X’s antipodal pair spans only one degree of freedom, so Fig. 5(b) only has a single axis. The color at a point g names the skill attaining the largest return gap $\max_z \langle g, u_z \rangle$, saturation gives that gap and fades to white at zero, and the black dot marks $g = 0$. J averages that gap over the prior (Eq. (12)). In the information panels, each point is a possible skill’s DSOM, at the vertices of a reachable polytope in Fig. 5 and at a specified channel in Fig. 7. The skill DSOM are shown explicitly in the inset bar chart. Arrows point from the capacity-achieving mixture q^* (black dot) to every possible skill’s DSOM, including those the capacity-achieving source gives zero weight. Support is a property of the source, not of reachability. The dashed line traces the KL level set at the capacity.

K.5 Worked example of centrality in information geometry

We illustrate the geometry of empowerment in Fig. 3a. To start, consider learning skills that start at state s_2 . From this state, there are three actions which deterministically transition to s_3 , s_4 , or s_5 . We consider learning three skills, so $z \in \{1, 2, 3\}$. The problem of optimizing mutual information at state s_2 corresponds to learning the policy $\pi(a | s = s_2, z)$ so that the skill index z conveys as many bits as possible about the next state. A poor choice for this policy would be the random policy, wherein each skill selects randomly among the three actions. Each of these three skills induces the same distribution over future states s_3, s_4, s_5 , conveying zero bits of information. We visualize the induced state occupancies in

Fig. 3a. Conversely, if we assign each skill a unique action (e.g., $z = 1$ always takes the leftmost action), then the state distributions the skills induce become markedly different. As visualized in Fig. 3a, this choice of skills covers a much larger region.

In state s_1 , the transitions are stochastic. If we similarly learned a set of skills to maximize mutual information, they would arrive at occupancy measures denoted by the orange dots in Fig. 3a. We can now interpret this example through the lens of empowerment. Is the empowerment $\mathcal{E}_{\text{pot}}(s)$ higher in state s_1 or s_2 ? Looking at Fig. 3a, we see that the skills at state s_2 are more “spread out”; formally, they convey more bits to the future. Thus, an empowerment-maximizing policy would prefer to navigate towards and stay at state s_2 .

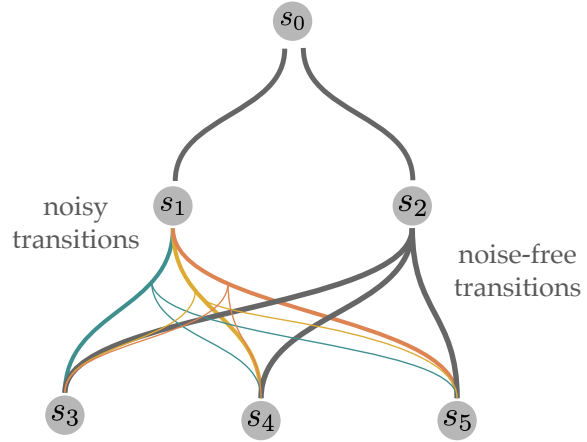


Figure 13: Worked Example of Empowerment.